

Digital Image Processing

Lect-1

Intro & Affiliations

Muhammad Usman Akram, PhD Computer Engineering

Area of research: Analysis of medical images/signals using Image/Signal Processing and Machine Learning Techniques

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RESEARCH AND INNOVATION
IN SCIENCE ENGINEERING AND TECHNOLOGY

www.biomisa.org/usman
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Image Processing

Image Analysis

Machine Learning

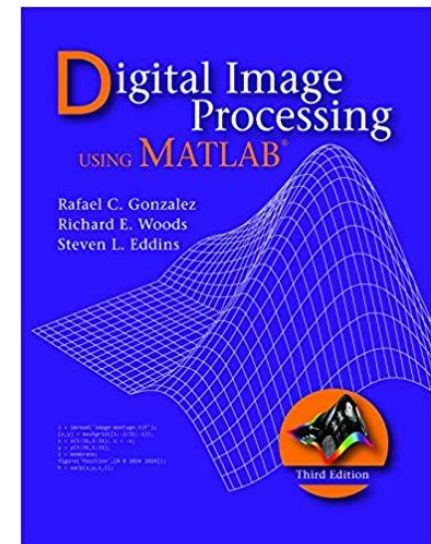
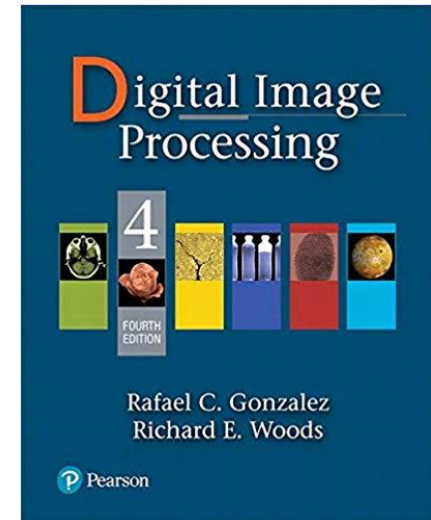
Computer Vision

Deep Learning



Text Book & References:

- Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, 4rd Edition, 2018 (available from local market)
- Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing using MATLAB*, 3rd Edition, 2020 (available from local market) (available from local market)
- Class slides & selected research papers to be shared by the instructor



Course Information

- **Course Material**

- Lectures slides, assignments (computer/written), solutions to problems, projects, and announcements will be uploaded on LMS.

Grading Policy

Exam:	2 Sessional and 1 Final			
Home work:	3 graded Assignments, 02-04 home tasks (non graded)			
Lab reports:	12-13 reports, 01 open Lab, 01 Lab Final			
Design reports:	1 Design report and 2 presentations based on Semester Project (First 3 Min Brief, Second 10 Min complete project presentation)			
Quizzes:	6-8 Quizzes			
Grading:	Theory (67%)		Lab (33%)	
	2 One Hour Tests (OHTs):	30%	Lab Work/Tasks (Every week)	45%
			Open Lab	10%
	Quizzes:	10%	Lab Final	10%
	Assignments	10%	Home Tasks	05%
	Final Exam	50%	Final Project	30%



Course Contents

- Introduction to Image processing (Chapter – 1, 2)
- Image processing Fundamentals (Chapter-2)
- Image Enhancement (Spatial & Frequency Domain) (Chapter – 3, 4)
- Morphological operations (Chapter – 9)
- Segmentation (Chapter – 10, Online material, David Forsyth)
- Texture analysis (Chapter – 11, Online material, David Forsyth)
- Image Pyramids & Wavelets (Chapter – 07, Online material, David Forsyth)
- Image representation and description (Chapter – 11)
- Introduction to Machine Learning and Convolutional Neural Networks (Chapter – 12 & Stanford course on CNN)

Lab Breakdown

Lab 01	Installation & Introduction to Python and OpenCV
Lab 02	Basic Image Processing (Group formation for semester project)
Lab 03	Connected Component Analysis (Assignment-1: Using Connect Component for Image Analysis)
Lab 04	Transformation Operations
Lab 05	Histogram Equalization and other transformations (Project Assignment)
Lab 06	Spatial Filtering (3 Minute presentation on project Idea)
Lab 07	Edge Detection and Segmentation (Assignment-2 Use of filtering, edge detection and segmentation for image analysis)
Lab 08	Open Lab
Lab 09	Morphological Operations (Seminar on using GitHub and Co-Lab)
Lab 10	Color Processing & Clustering
Lab 11	Image Classification using CNN
Lab 12	Object detection and localization (Assignment-3: object localization in medical images)
Lab 13	Image segmentation using deep learning
Lab 14	Frequency Analysis
Lab 15	Lab Final (Project Presentation and Submission)

Course Learning Outcome

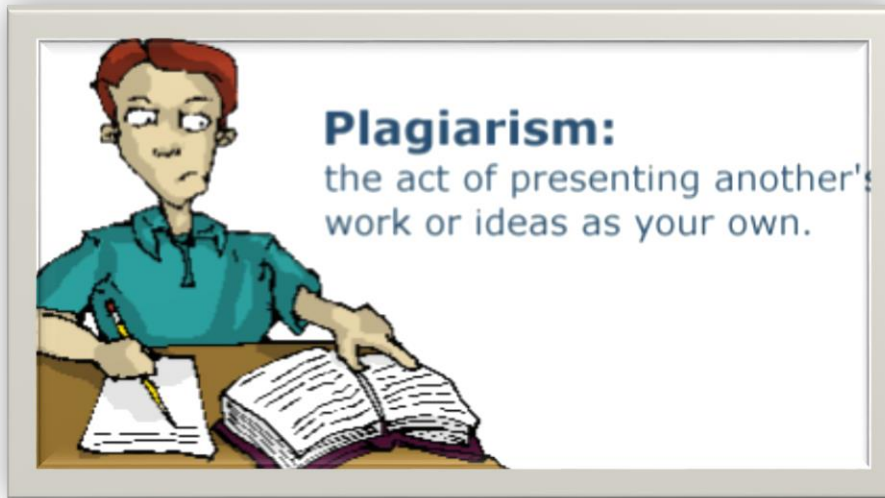
Course Learning Outcome (CLOs)		PLOs	Learning Level	Assessments
CLO 1	Understanding the fundamentals and basic concepts of image processing related to image enhancement, filtering and segmentation etc	PLO 1	C2	Q1, Q2, A1, OHT-1, Final 1-question
CLO 2	Performing different mathematical transformations, histogram based operations and filtering concepts for solving image enhancement and feature extraction problems	PLO 2	C3	Q3, Q4, A2, OHT2, Final 1-question
CLO 3	Combining the concepts of image processing with machine learning to analyze and design decision support systems for image processing based applications	PLO 3	C4	Q6, A3, Final 1-question
CLO 4	Learning the use of Python and OpenCV to implement basic image processing algorithms and to solve real life and open ended problems	PLO 5	P4	Labs, Open Labs, Project, Presentation

CODE OF ETHICS

- All students must come to class on time (Attendance will be taken in first 5 to 10 mins)
- Students should remain attentive during class and avoid use of Mobile phone, Laptops or any gadgets
- Obedience to all laws, discipline code, rules and community norms
- Respect peers, faculty and staff through actions and speech
- Bring writing material and books
- Class participation is encouraged

Policies

- No extensions in assignment deadlines.
- Quizzes will be unannounced.
- Final Exam will be OPEN book
- No Attendance and marking of lab tasks if late more than 15 min
- Never cheat.
 - “Better fail NOW or else will fail somewhere LATER in life”
- Plagiarism will also have strict penalties.



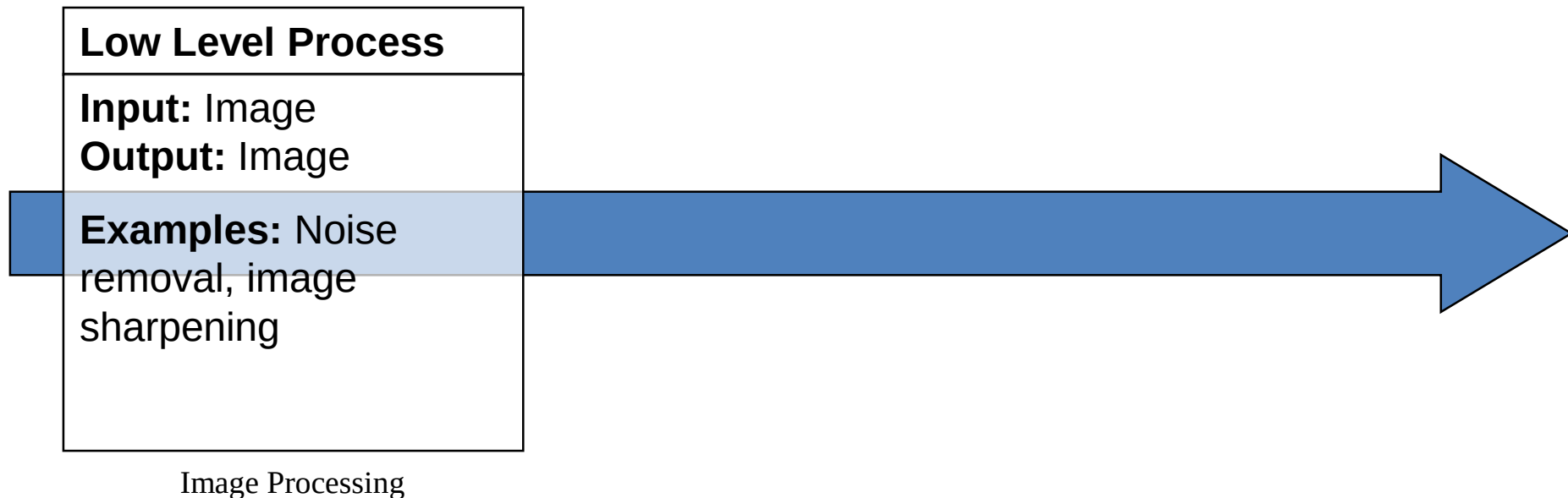
Courtesy Dr. Khawar

Adapted from *What is Plagiarism* PowerPoint
<http://mciu.org/~spjvweb/plagiarism.ppt>

Image Processing & Machine Vision

◆ From Image Processing to Machine Vision:

- low, mid and high-level processes



Example: Low Level Processing

Photo restoration



Damaged Image

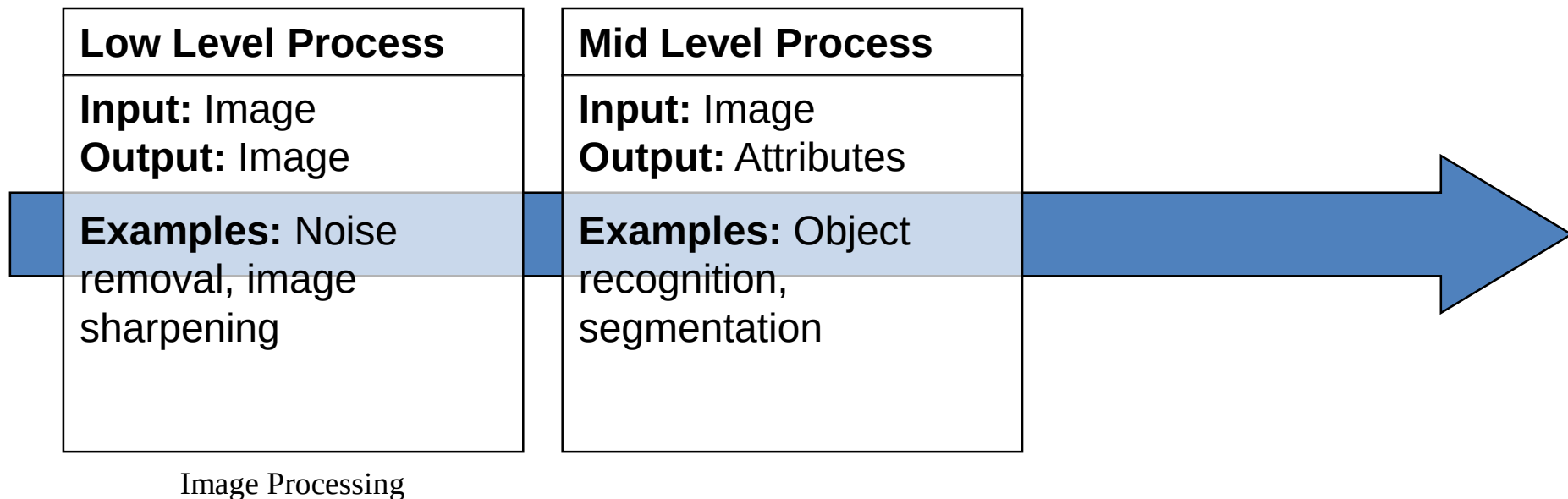


Restored Image

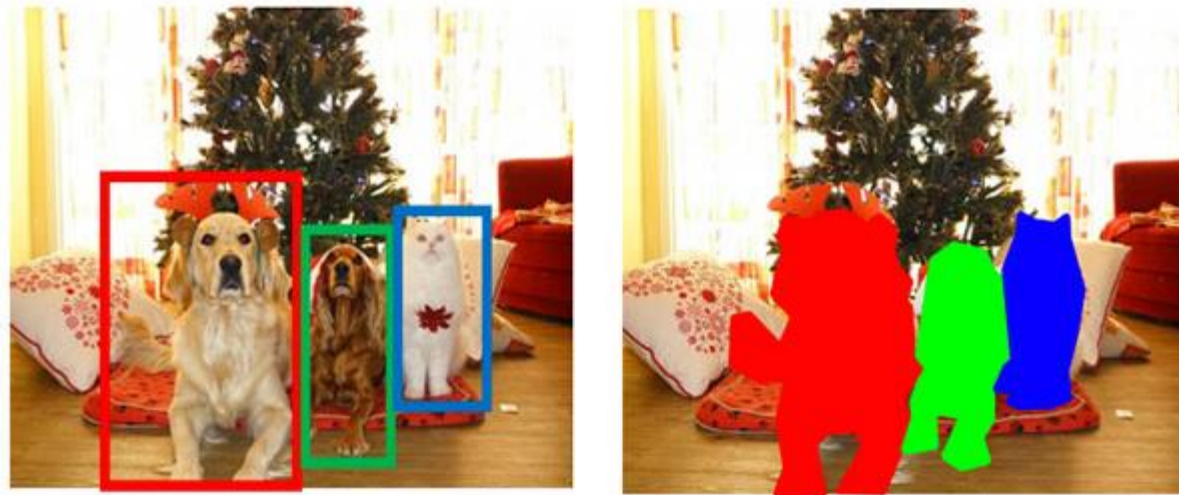
Image Processing & Machine Vision

◆ From Image Processing to Machine Vision:

- low, mid and high-level processes



Example: Mid Level Processing

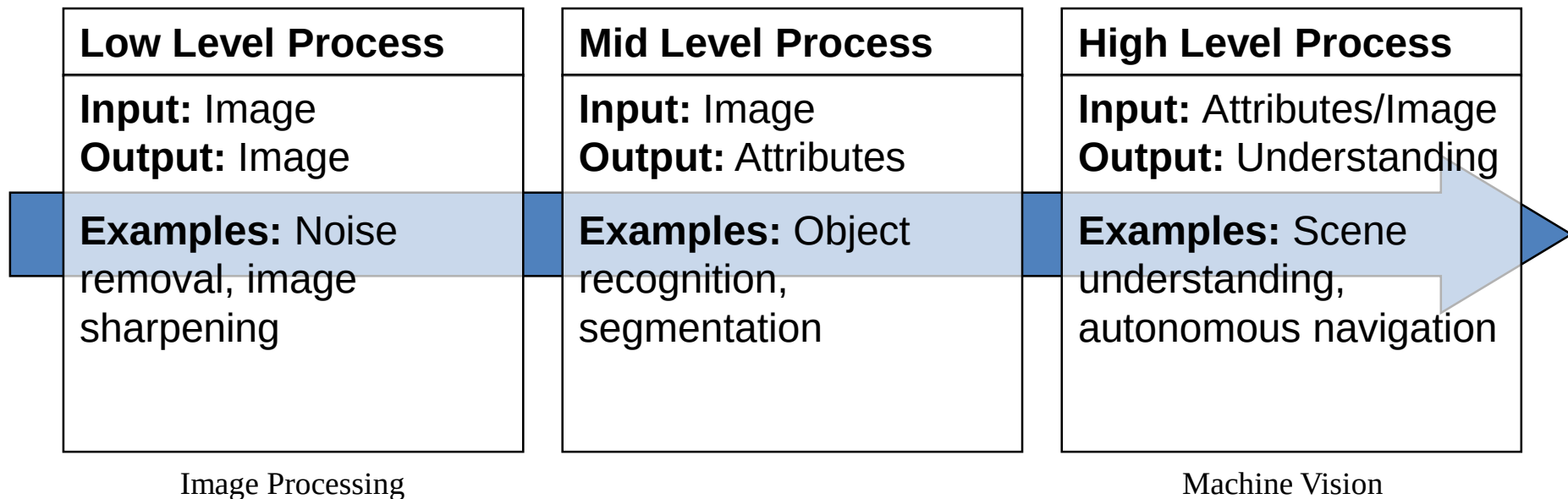


Segmentation of image into regions

Image Processing & Machine Vision

◆ From Image Processing to Machine Vision:

- low, mid and high-level processes



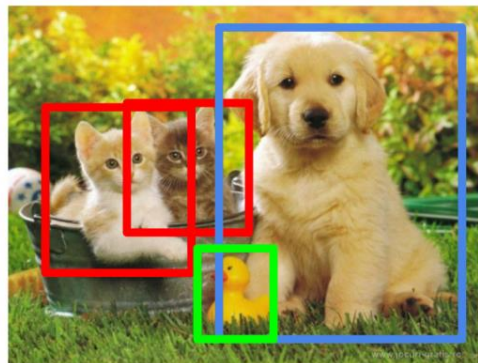
Example: High Level Processing



CAT



CAT



CAT, DOG, DUCK



CAT, DOG, DUCK

Object Classification

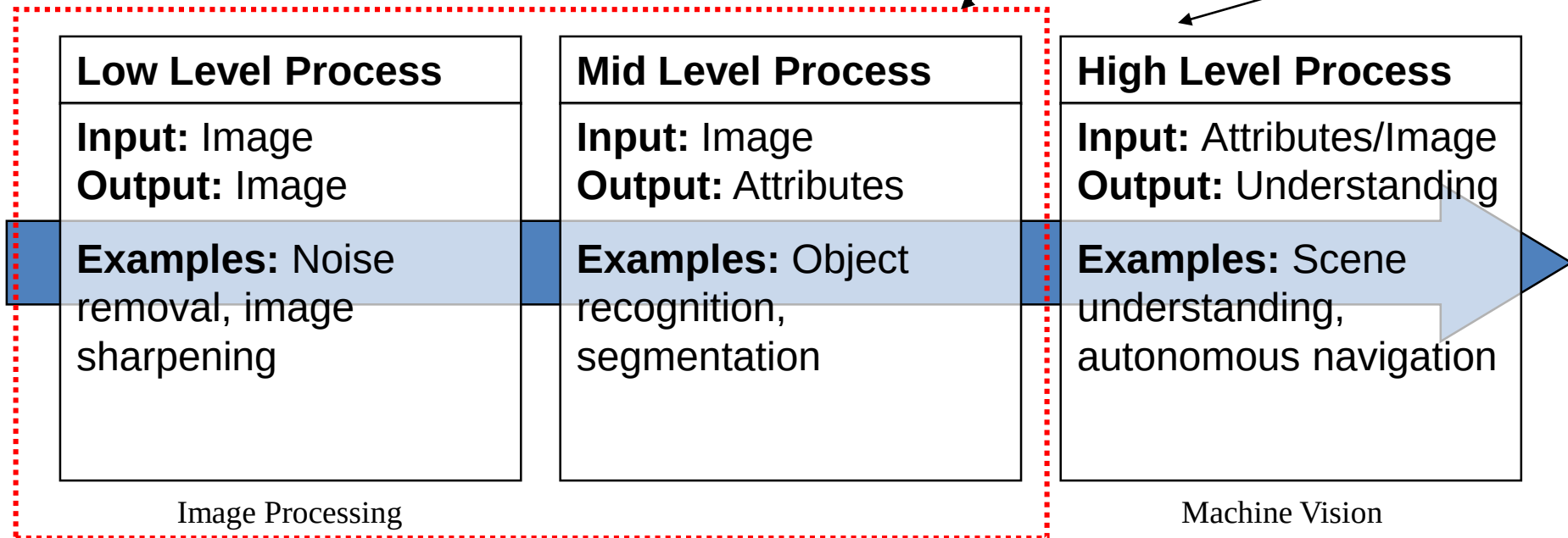
Image Processing & Machine Vision

◆ From Image Processing to Machine Vision:

- low, mid and high-level processes

In this course

Some of this as well



Why Image Processing?

- Images and video are everywhere!



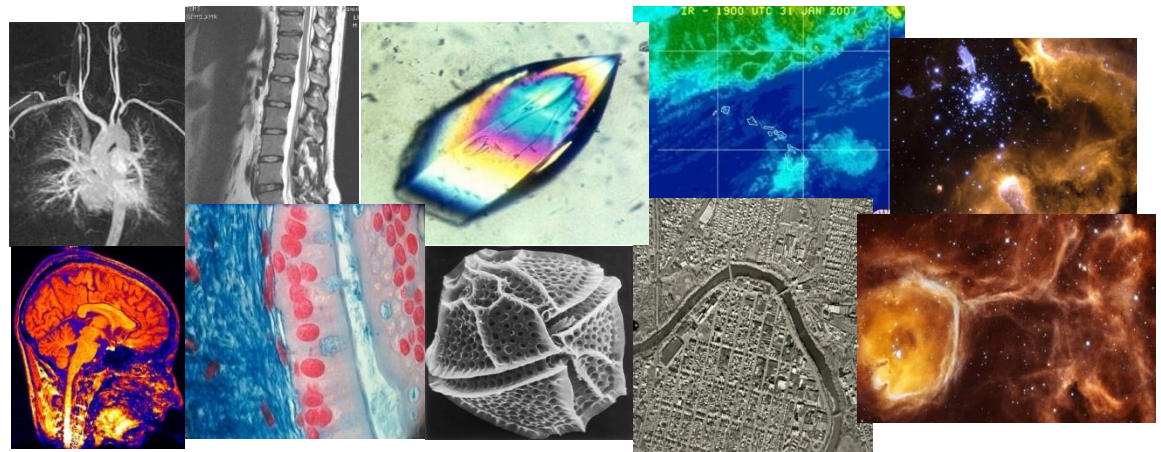
Personal photo albums



Movies, news, sports



Surveillance and security



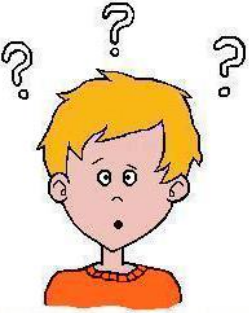
Medical and scientific images

Summary of Applications

Problem Domain	Application	Input Pattern	Output Class
Document Image Analysis	Optical Character Recognition	Document Image	Characters/words
Document Classification	Internet search	Text Document	Semantic categories
Document Classification	Junk mail filtering	Email	Junk/Non-Junk
Multimedia retrieval	Internet search	Video clip	Video genres
Speech Recognition	Telephone directory assistance	Speech waveform	Spoken words
Natural Language Processing	Information extraction	Sentence	Parts of Speech
Biometric Recognition	Personal identification	Face, finger print, Iris	Authorized users for access control
Medical	Computer aided diagnosis	Microscopic Image	Healthy/cancerous cell
Military	Automatic target recognition	Infrared image	Target type
Industrial automation	Fruit sorting	Images taken on conveyor belt	Grade of quality
Bioinformatics	Sequence analysis	DNA sequence	Known types of genes

Image Sources

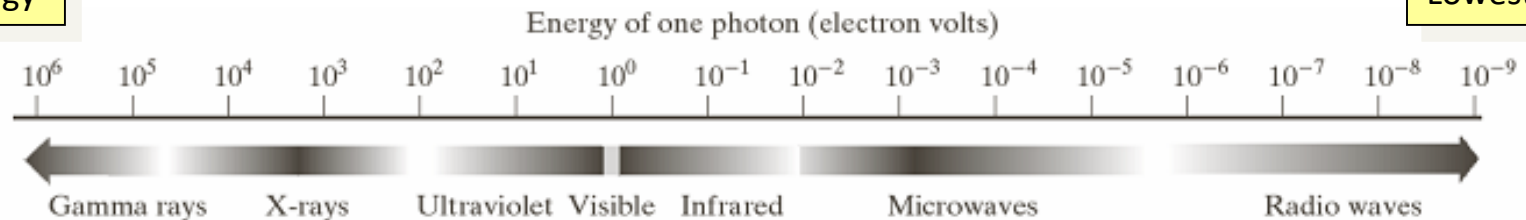
- **Electromagnetic (EM) band imaging**
 - Gamma ray band images
 - X-ray band images
 - Ultra violet band images
 - Visual light and infra-red images
 - Images based on micro waves or radio
- **Non-EM band imaging**
 - Acoustic and ultrasonic images
 - Electron microscopy
 - Computer generated images (synthetic)



Light & EM Spectrum

- EM Waves
 - A stream of mass less particles each travelling in a wave like pattern, moving at the speed of light and contains a certain bundle of energy
 - The electromagnetic spectrum is split up in to bands according to the energy per photon

Highest Energy



Lowest Energy

Light & EM Spectrum

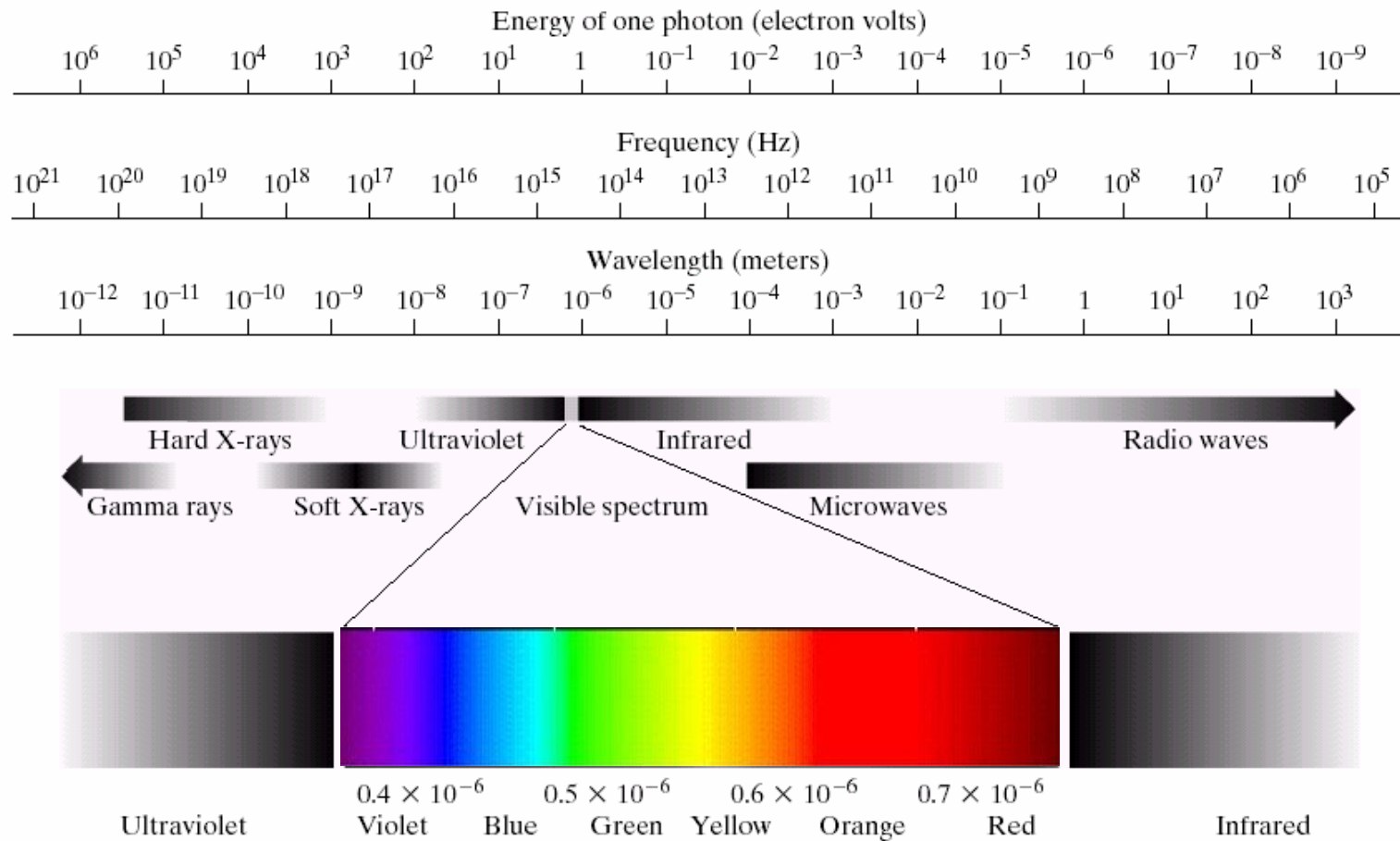
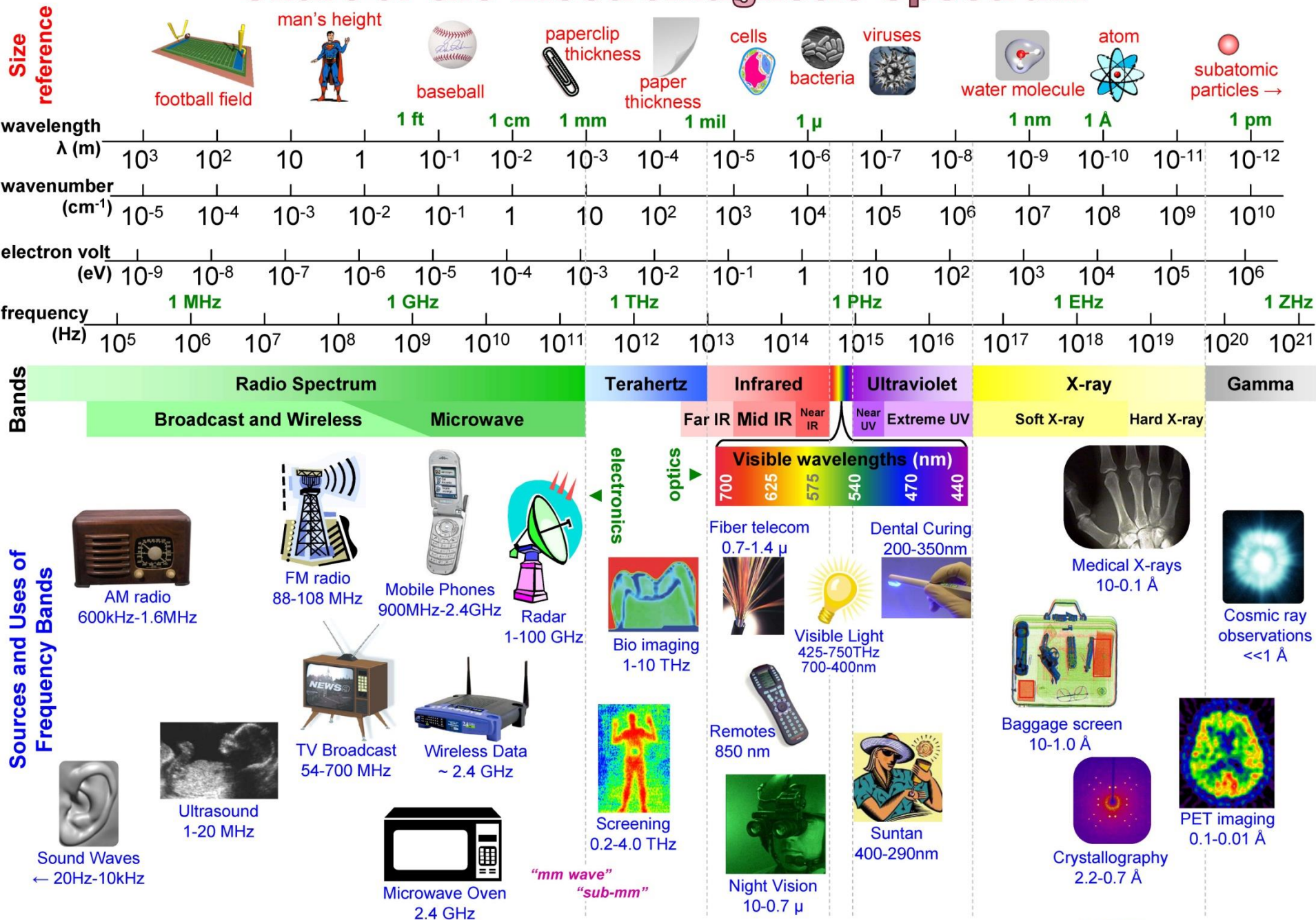


Chart of the Electromagnetic Spectrum



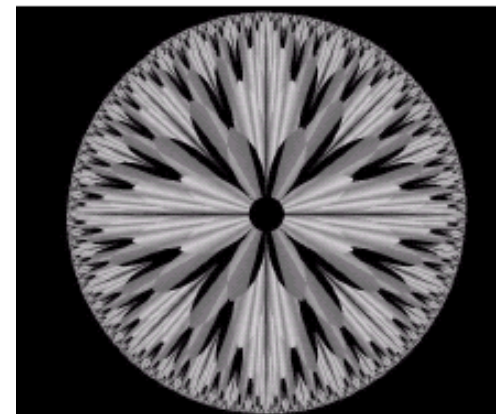
$$\lambda = 3 \times 10^8 / \text{freq} = 1 / (\text{wn} \times 100) = 1.24 \times 10^{-6} / \text{eV}$$

Examples: Imaging in EM bands

Spectral Band	Example
Gamma-Rays	Nuclear Medicine (Radioactive isotope injected in the patient)
X-Rays	Medical Diagnostic
Ultraviolet	Fluorescence microscopy
Visible & Infrared	Remote sensing, industrial inspection, ...
Microwave	Radar
Radio	Medicine – MRI

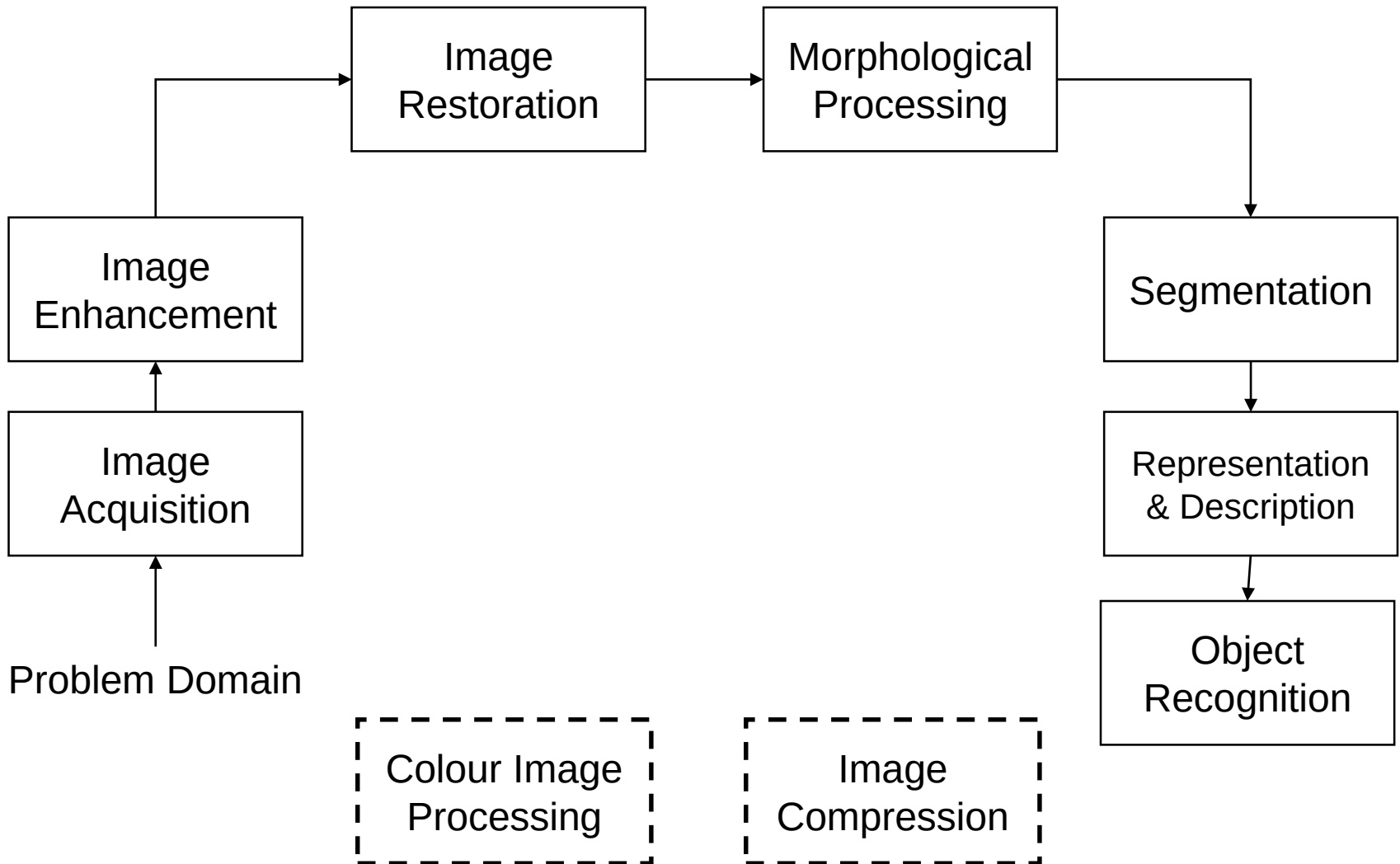
Examples: Imaging other Modalities

- Sound
 - Geological Applications – Oil and Gas Exploration
 - Medicine – Ultrasound Imaging
- Synthetic Images
 - Computer generated

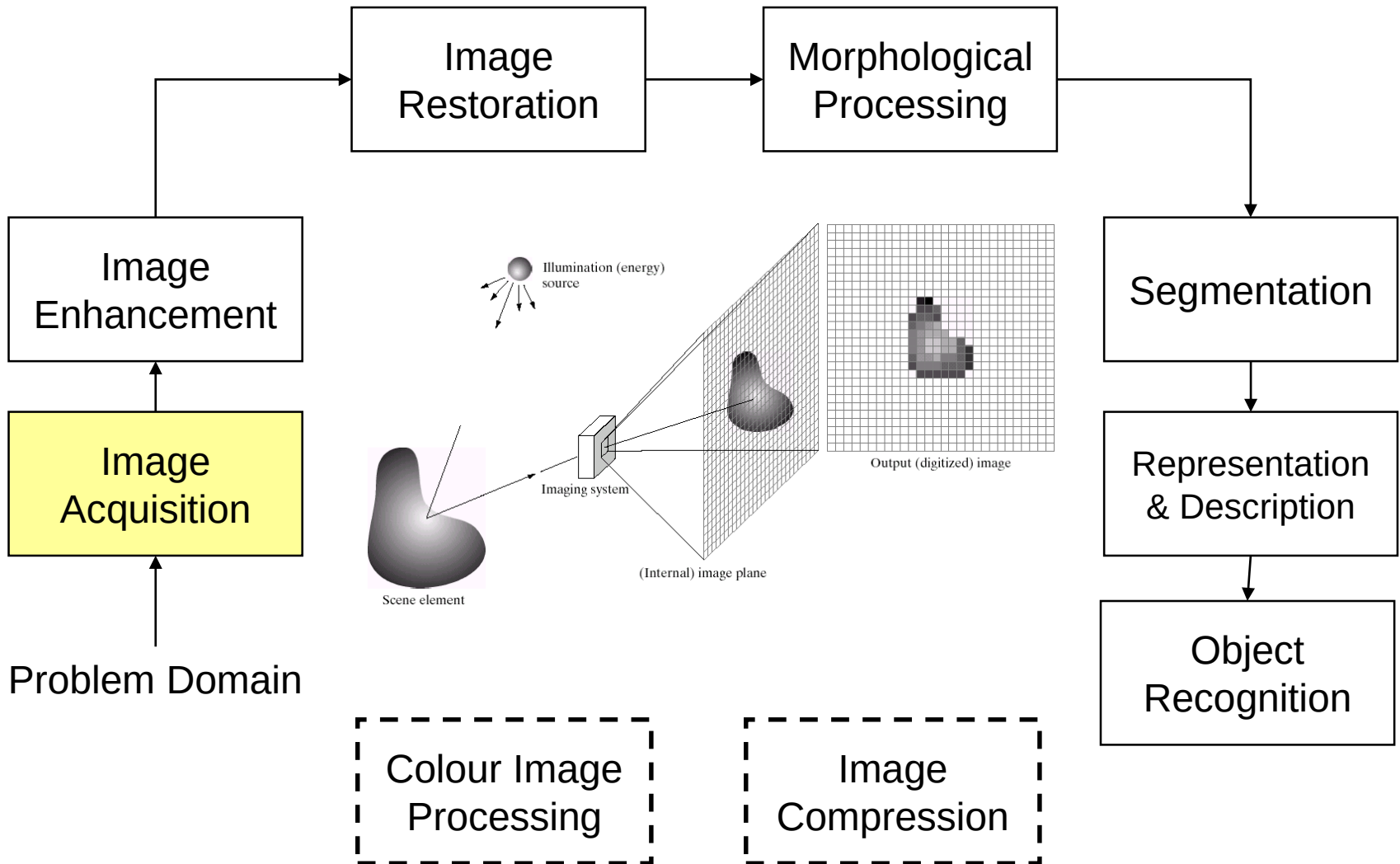


A synthetic image

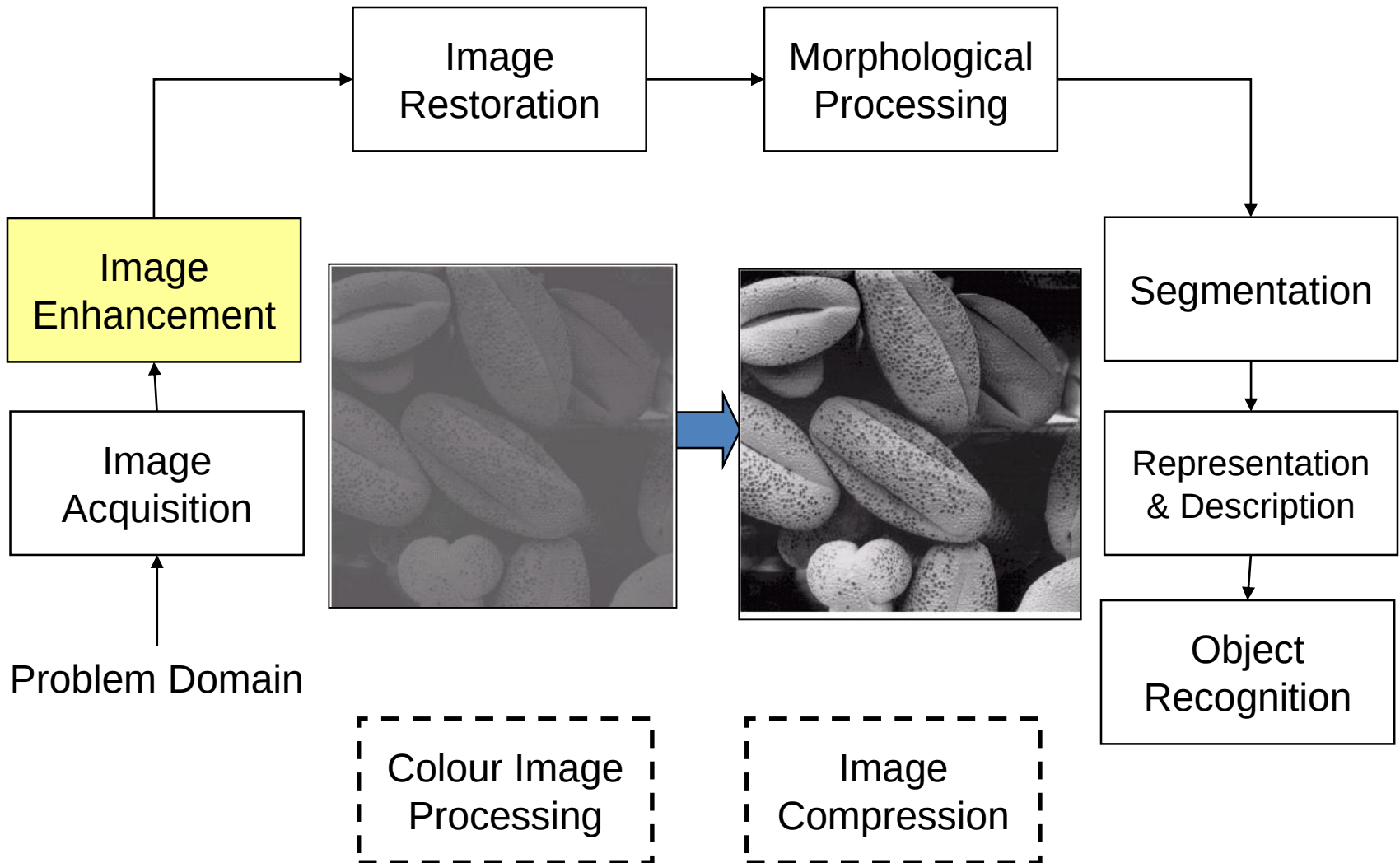
Key Stages in DIP



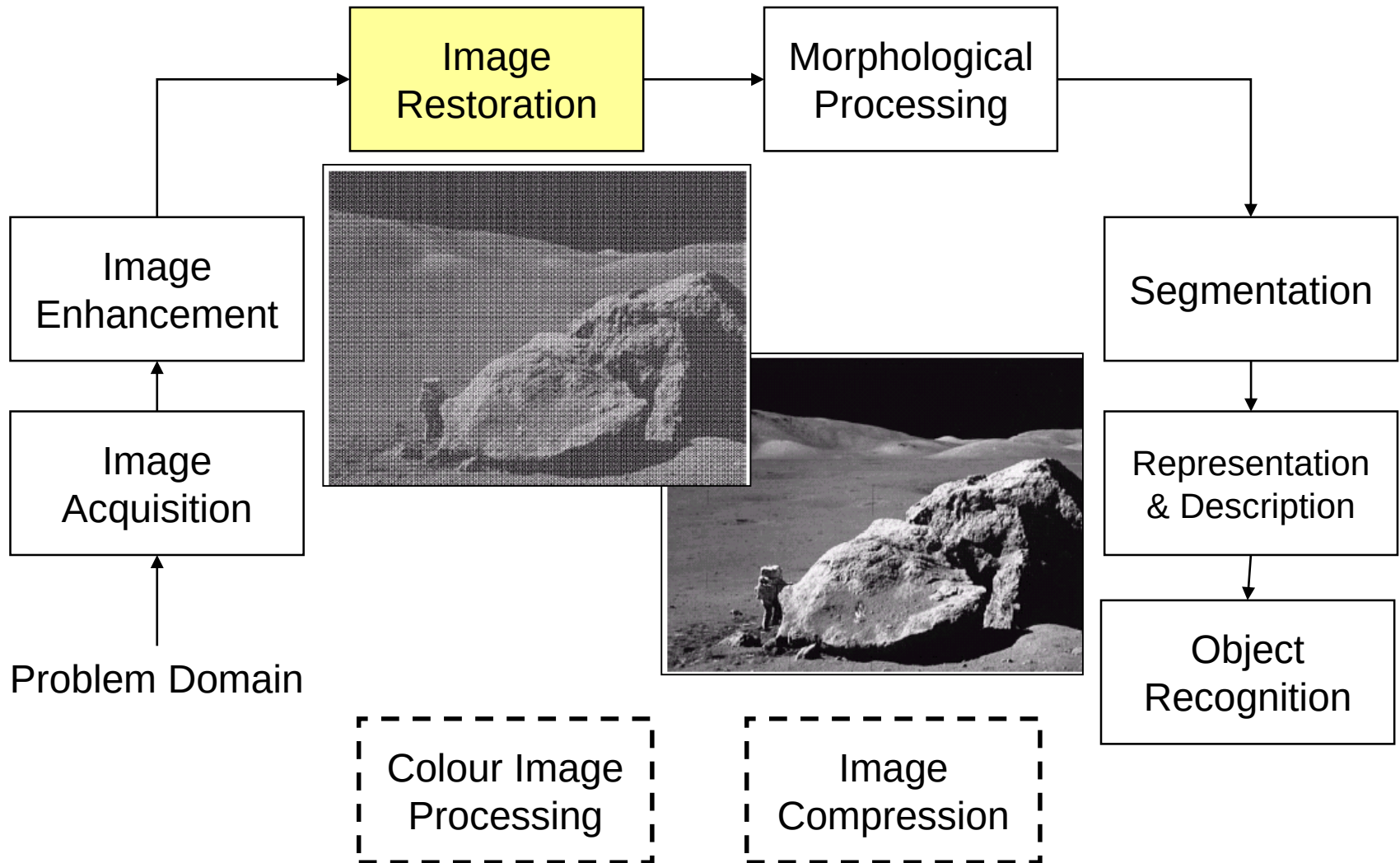
Key Stages in DIP



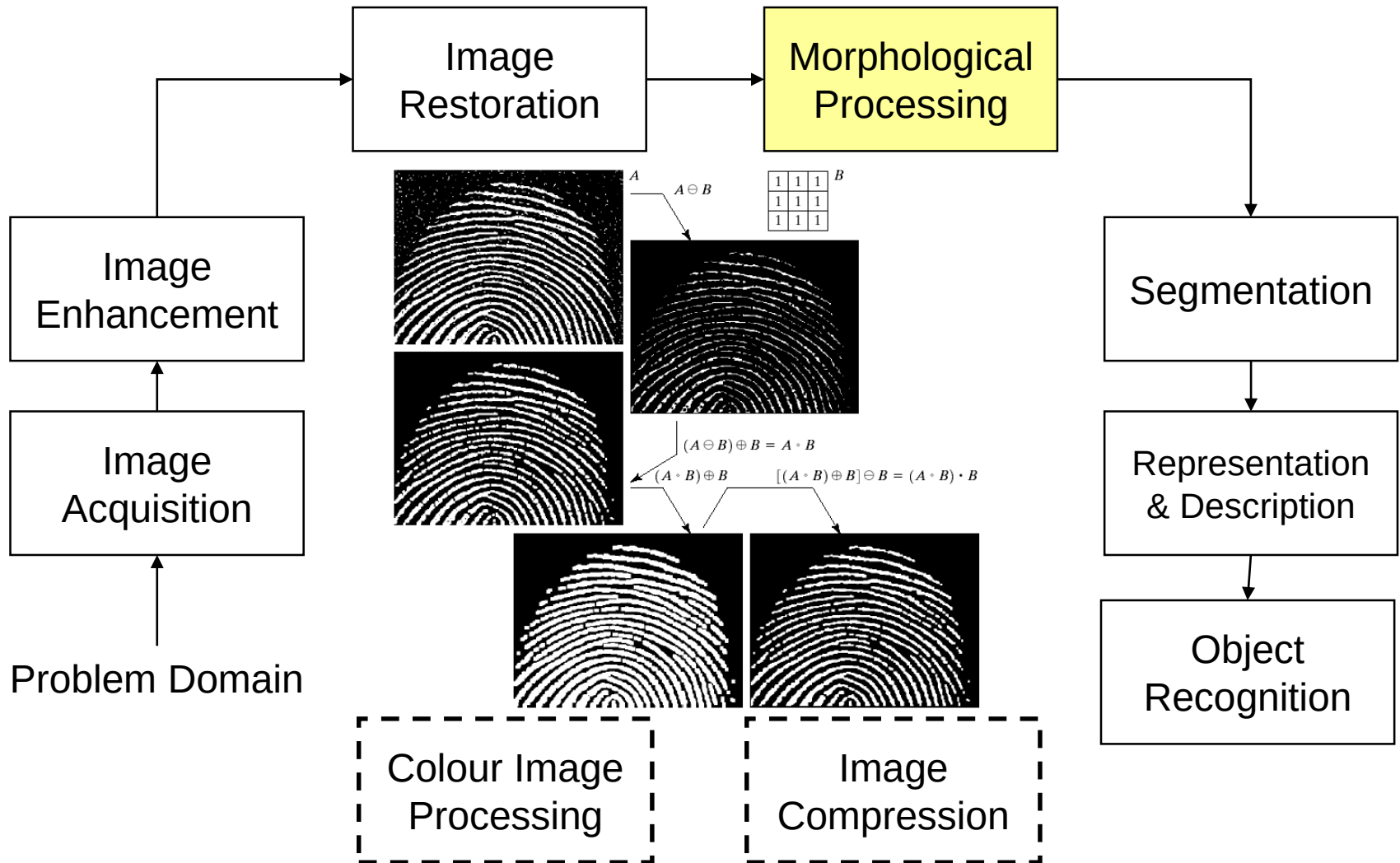
Key Stages in DIP



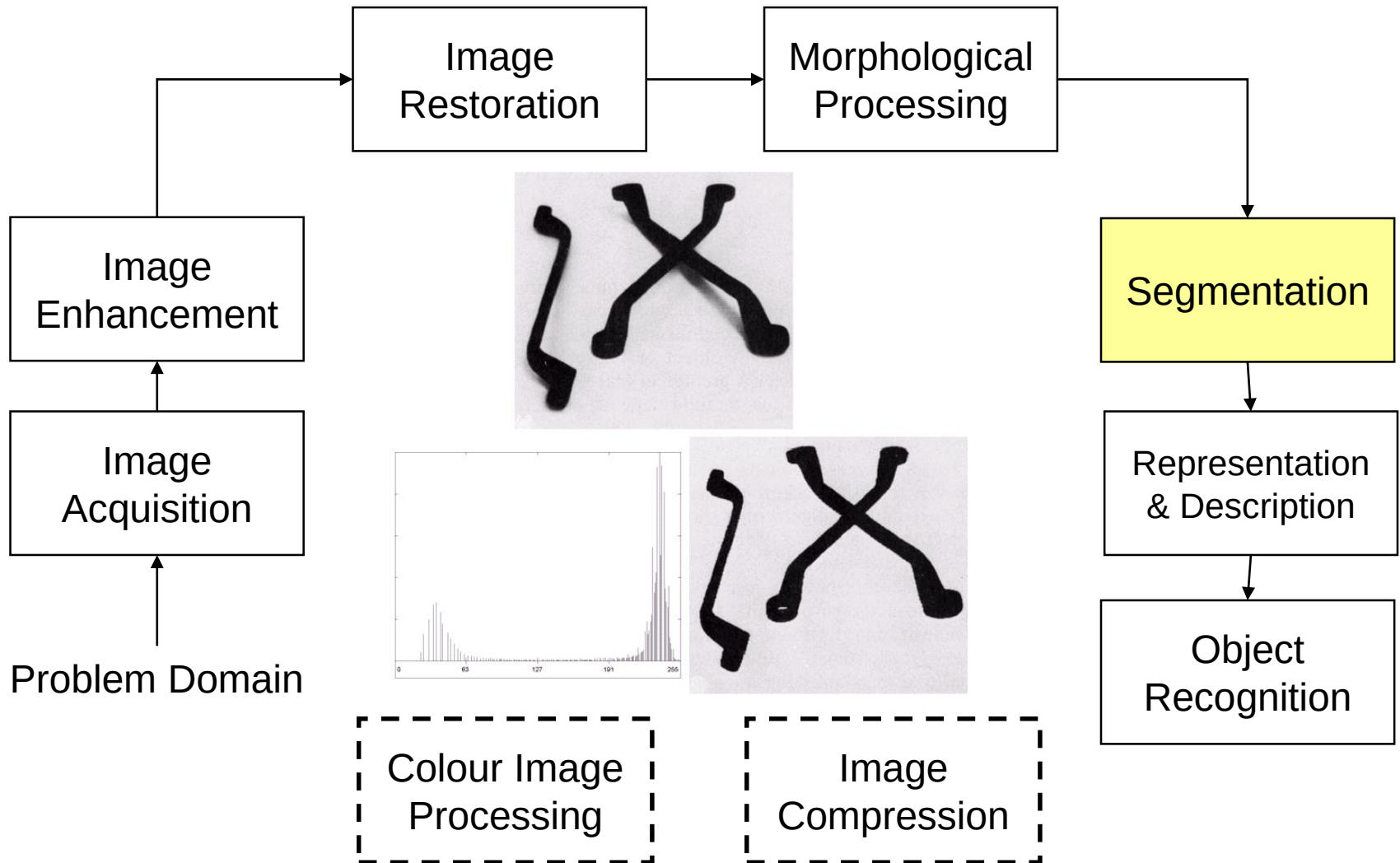
Key Stages in DIP



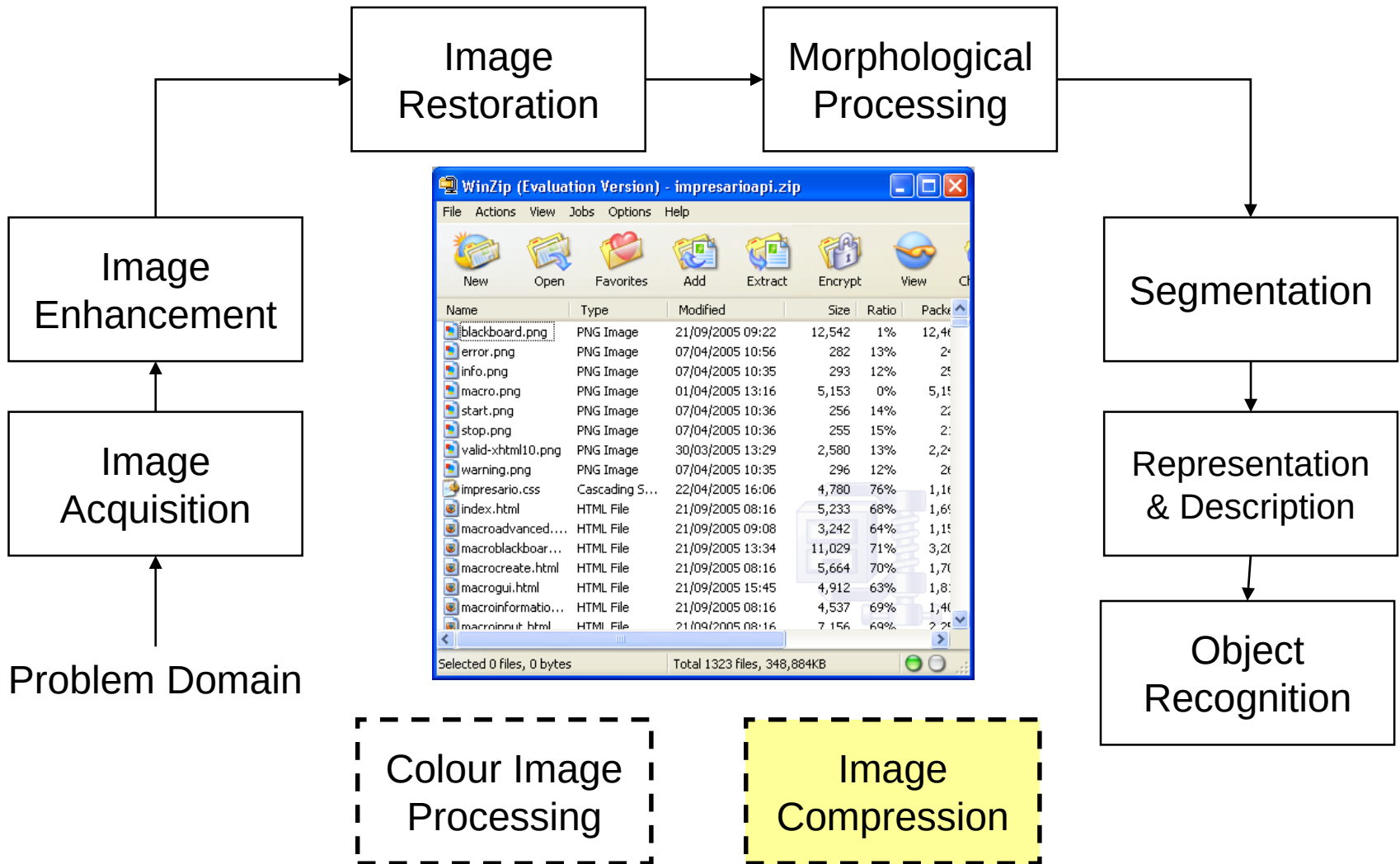
Key Stages in DIP



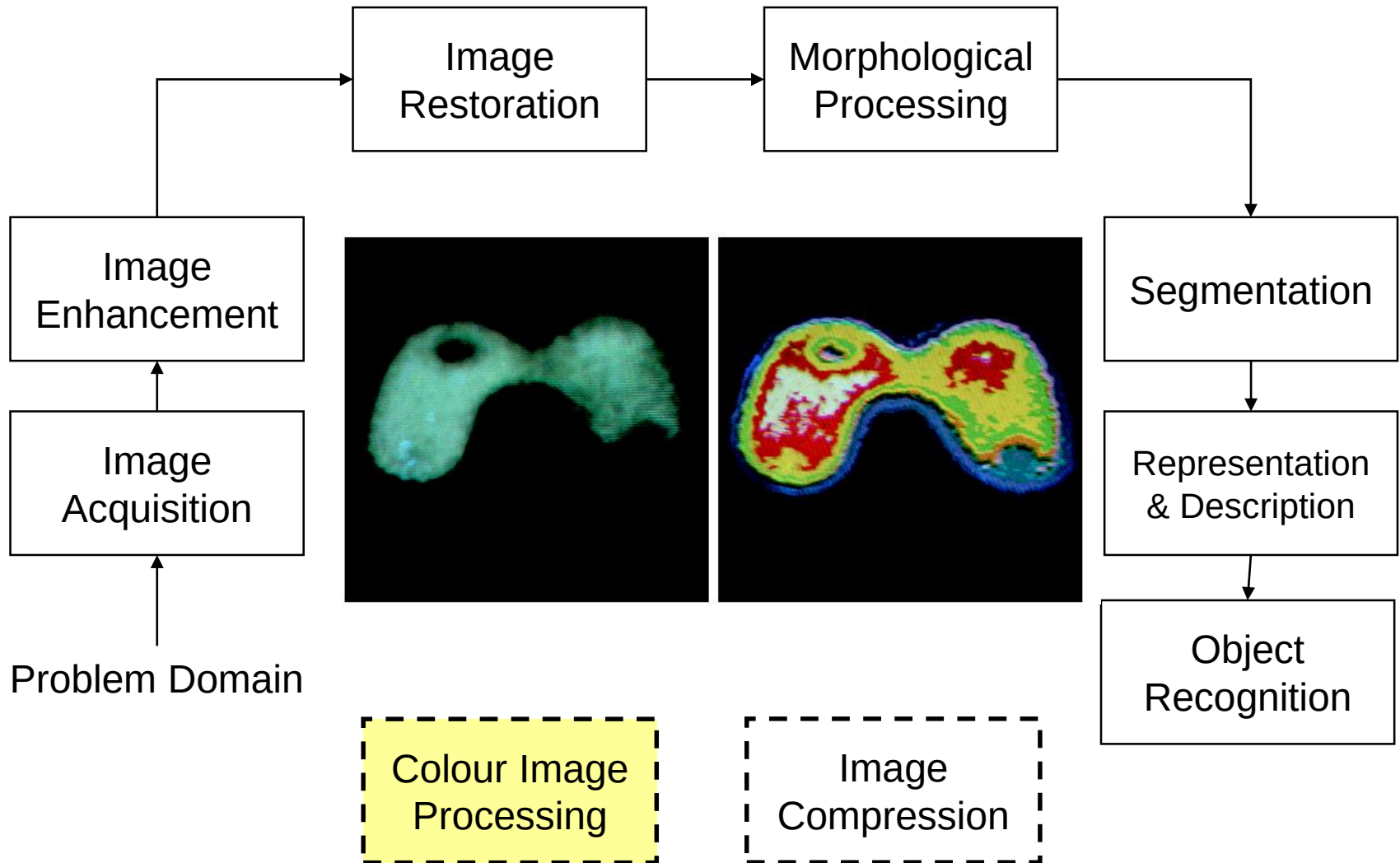
Key Stages in DIP



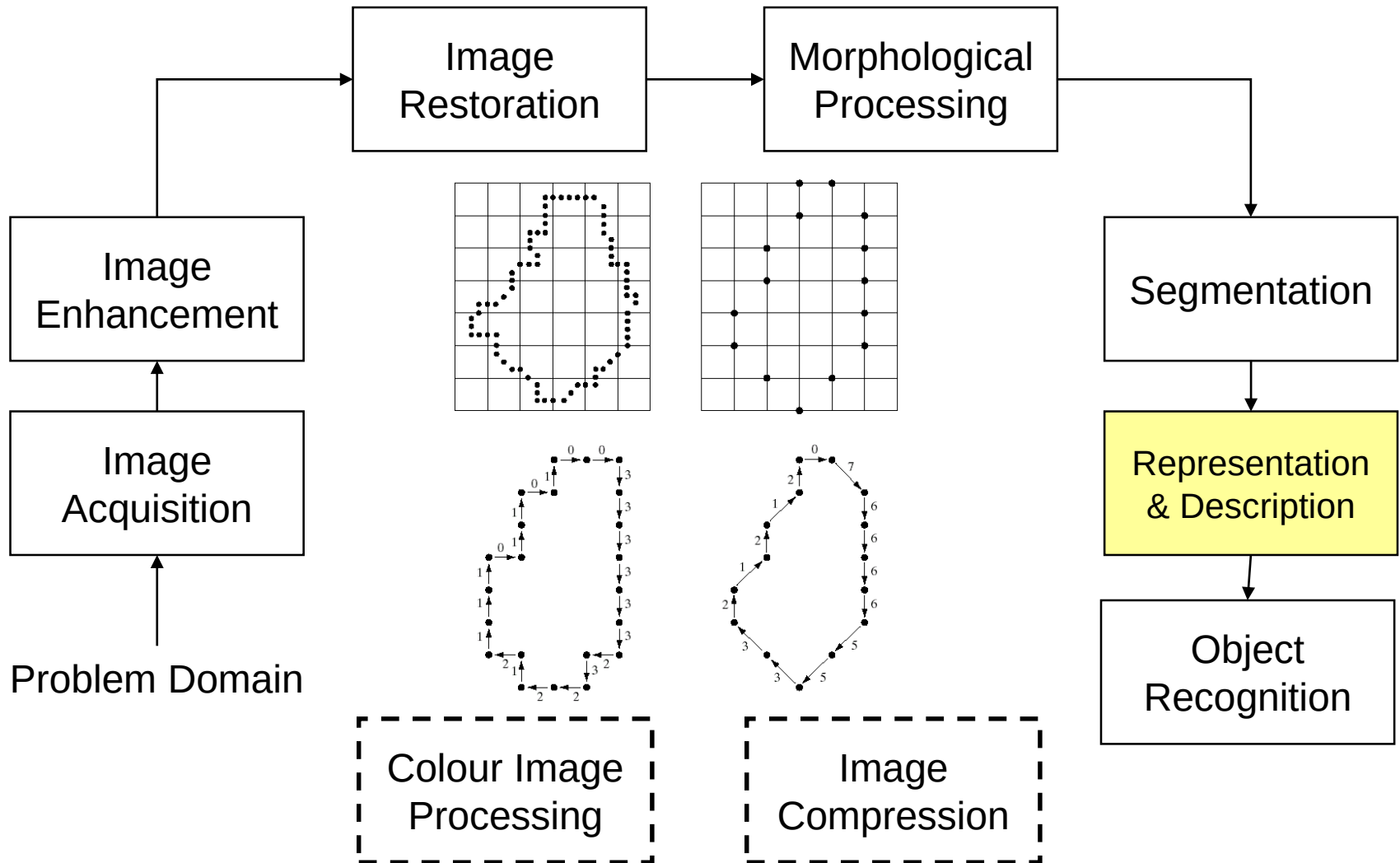
Key Stages in DIP



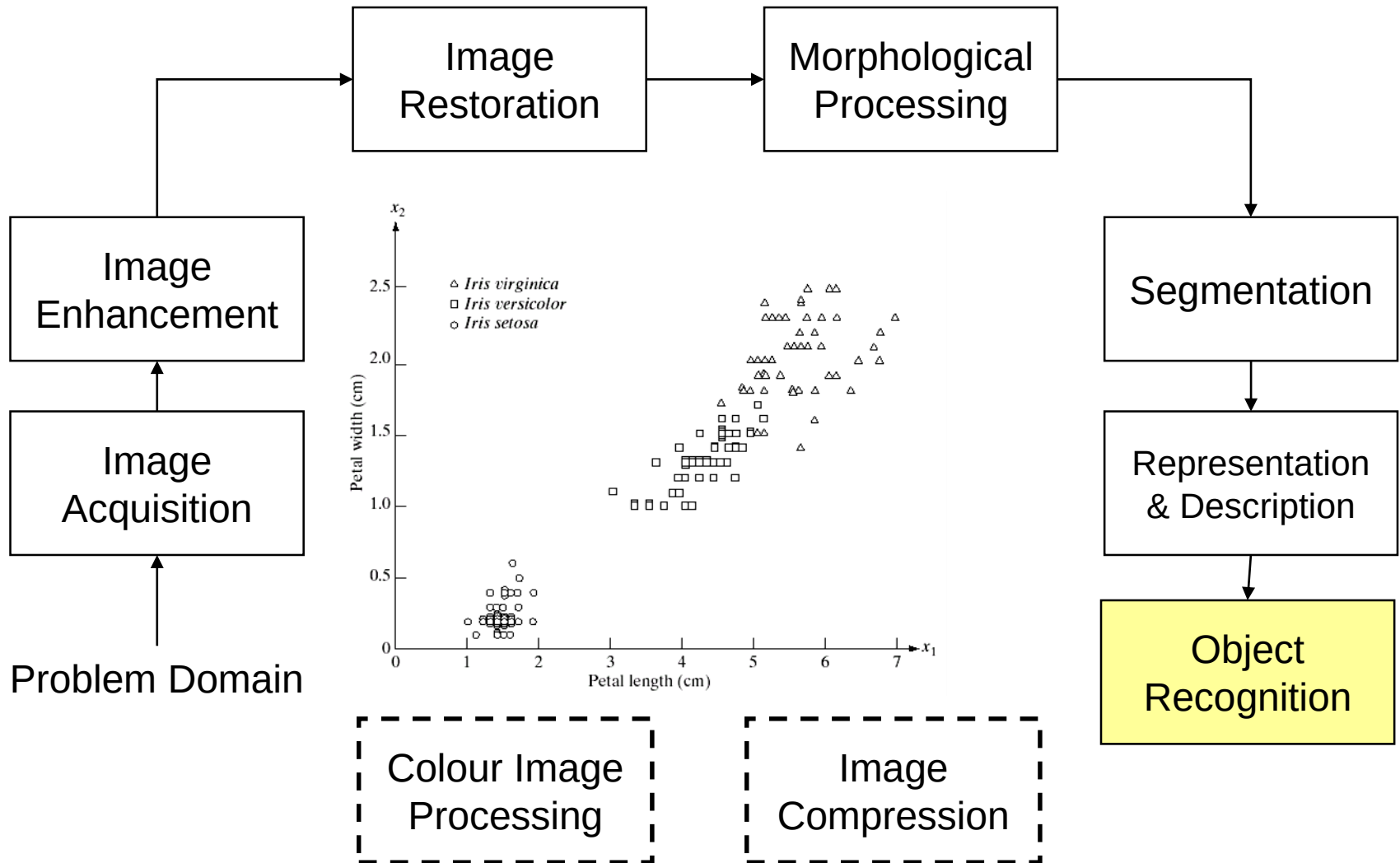
Key Stages in DIP



Key Stages in DIP



Key Stages in DIP



Digital Image Processing

Fundamentals

IMAGE FORMATION MODEL

- Image refers to a 2d light-intensity function, $f(x, y)$
- The amplitude of f at spatial coordinates (x, y) gives the intensity (brightness) of the image at that point.
- Light is a form of energy thus $f(x, y)$ must be nonzero and finite.

$$0 < f(x, y) < \infty.$$

IMAGE FORMATION MODEL

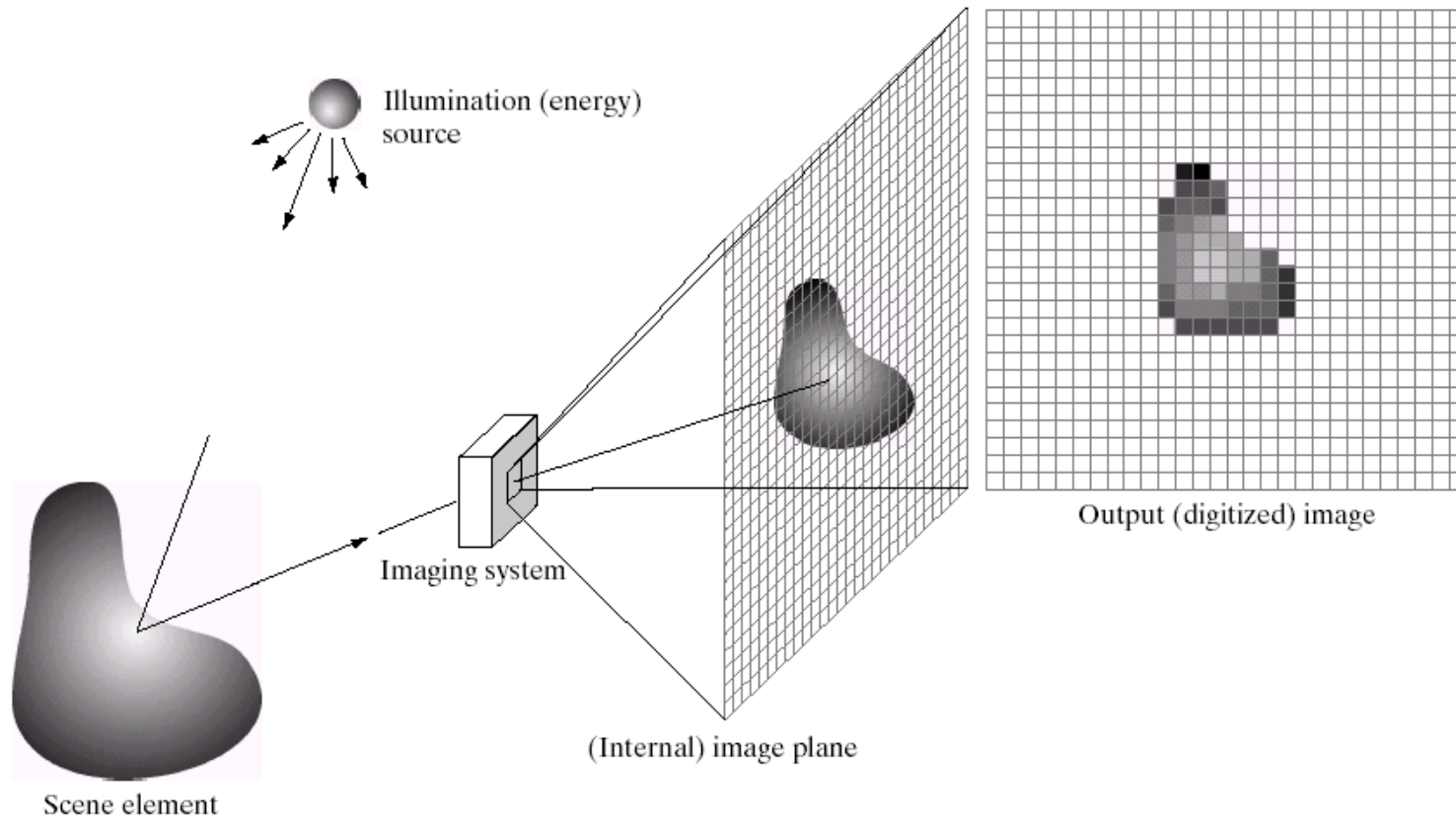
- The function $f(x, y)$ may be characterized by two components:
 - The amount of source light incident on the scene being viewed $\Rightarrow$ illumination.
 - The amount of light reflected by the objects in the scene $\Rightarrow$ reflectance.

$$f(x, y) = i(x, y)r(x, y)$$

$$0 < i(x, y) < \infty$$

$$0 < r(x, y) < 1.$$

Image Acquisition



a b c d e

FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Sampling and Quantization

- ◆ Sampling:

- Digitization of the spatial coordinates (x,y)

- ◆ Quantization:

- Digitization in amplitude (also known as gray level quantization)

Sampling and Quantization

◆ Quantization

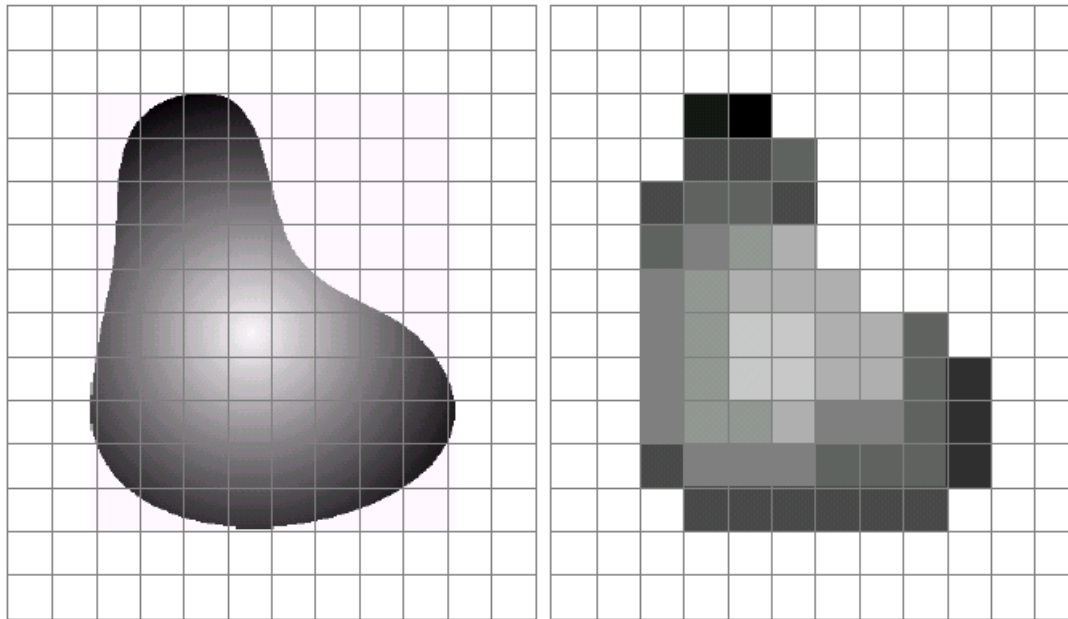
- 8 bit quantization: $2^8 = 256$ gray levels (0: black, 255: white)
- 1 bit quantization: 2 gray levels (0: black, 1: white) – binary

◆ Sampling

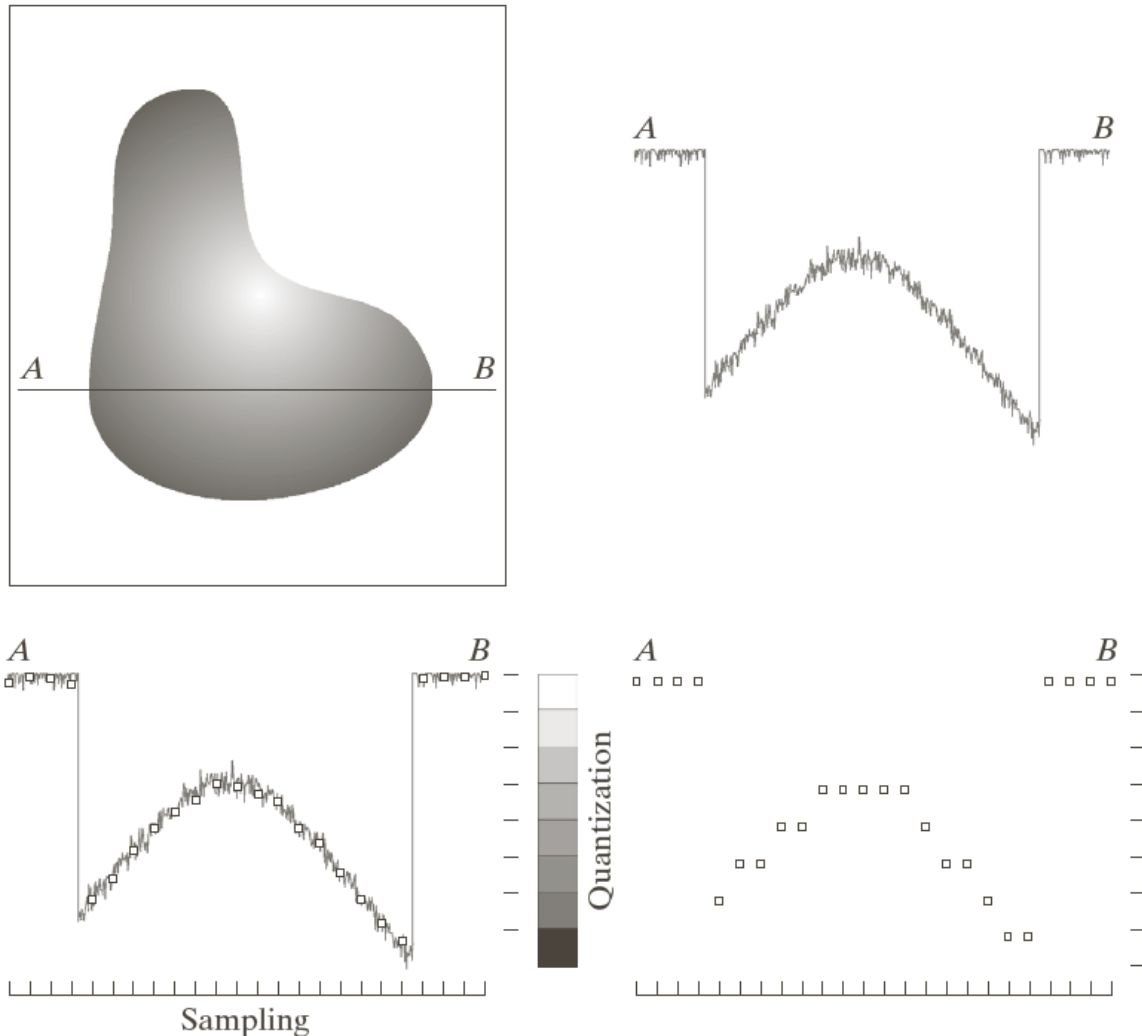
- Commonly used number of samples (resolution)
 - Digital still cameras: 640x480, 1024x1024, 4064 x 2704
 - Digital video cameras: 640x480 at 30 frames/second (fps)

Sampling and Quantization

- ◆ Digital Image is an approximation of a real world scene



Sampling and Quantization

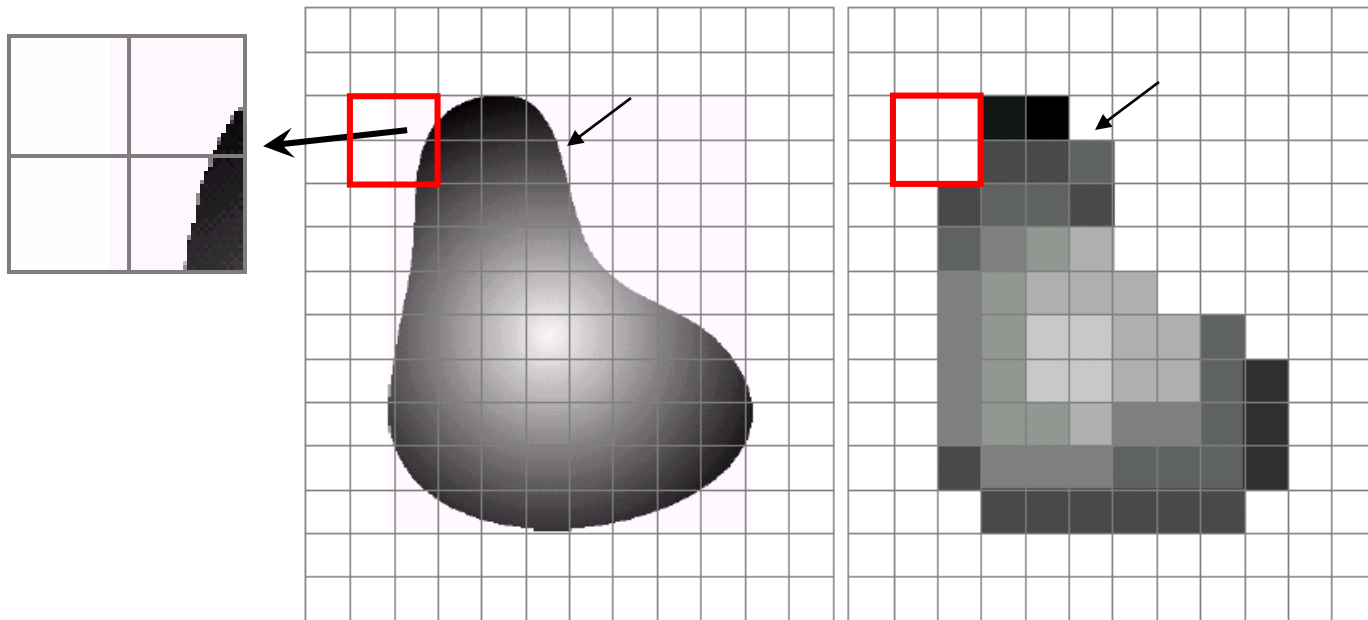


a	b
c	d

FIGURE 2.16
Generating a digital image.
(a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

Image Formation

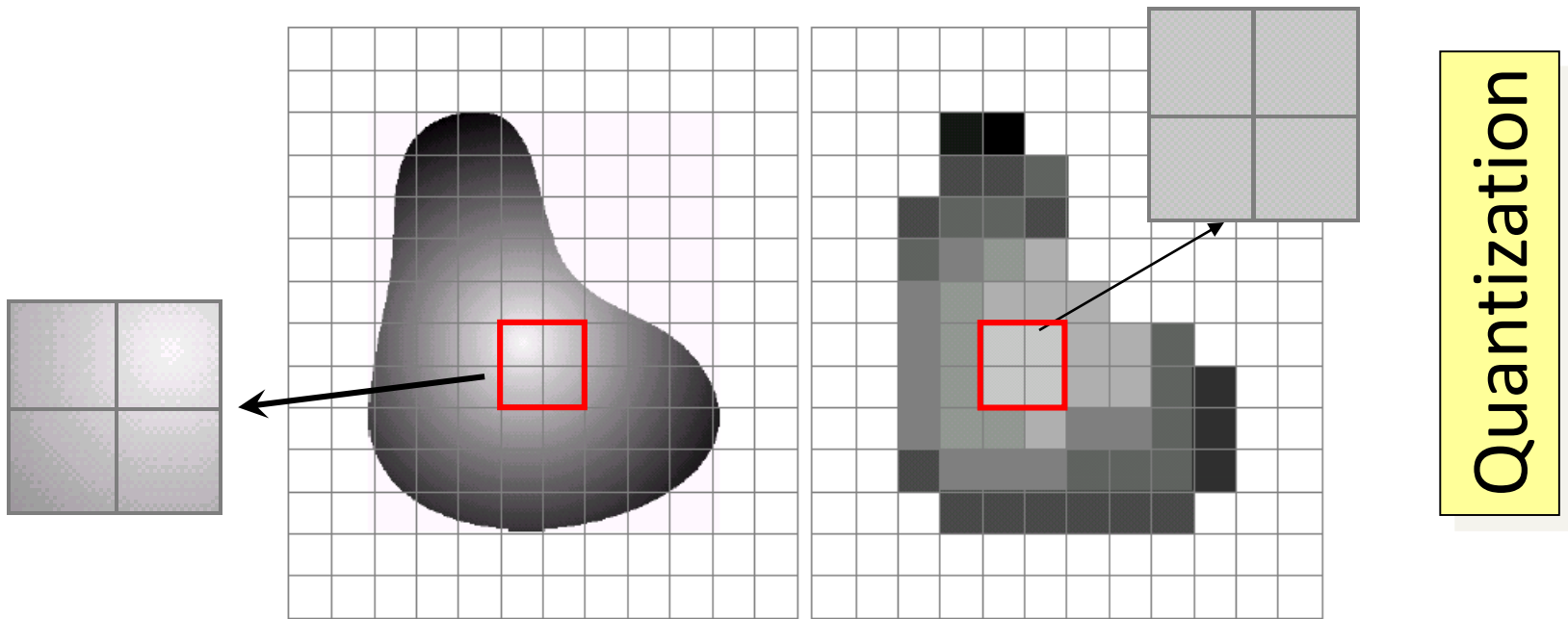
- ◆ Digital Image is an approximation of a real world scene



Sampling

Image Formation

- ◆ Digital Image is an approximation of a real world scene



GRAY LEVEL

- WE CALL THE INTENSITY OF A MONOCHROME IMAGE f AT COORDINATE (x, y) THE GRAY LEVEL (L) OF THE IMAGE AT THAT POINT.
- Thus, L lies in the range

$$L_{\min} \leq \ell \leq L_{\max}$$

- $L_{\min}$ is positive and $L_{\max}$ is finite.
- Gray scale = $[L_{\min}, L_{\max}]$
- Common practice, shift the interval to $[0, L]$: 0 = black, $L-1$ = white

Digital Image Representation

- ◆ Image Size

- Number of bits required to store an image

$$b = M \times N \times k$$

- Image having 2^k intensity levels
 - k – bit image
 - 256 intensity levels – 8 bit image

Image Size

TABLE 2.1

Number of storage bits for various values of N and k .

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Spatial & Gray Level Resolution

Spatial Resolution



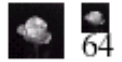
1024



512



256



128



64

32

Spatial Resolution



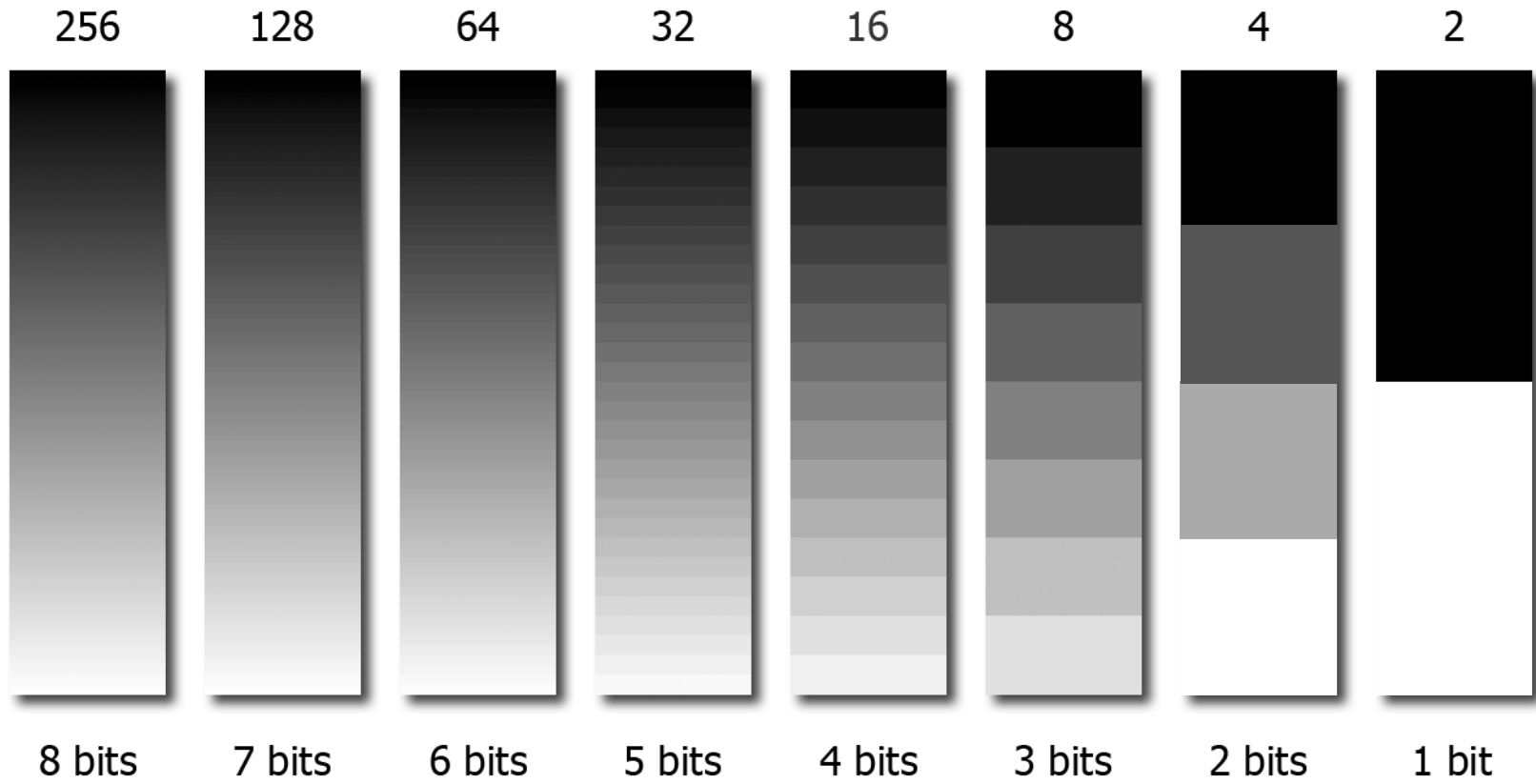
Intensity Level Resolution

- ◆ *Intensity level resolution* refers to the number of intensity levels used to represent the image
 - The more intensity levels used, the finer the level of detail in an image
 - Intensity level resolution is usually given in terms of the number of bits used to store each intensity level

Intensity Level Resolution

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Intensity Level Resolution



Intensity Level Resolution

256 grey levels (8 bits per pixel)



128 grey levels (7 bpp)



64 grey levels (6 bpp)



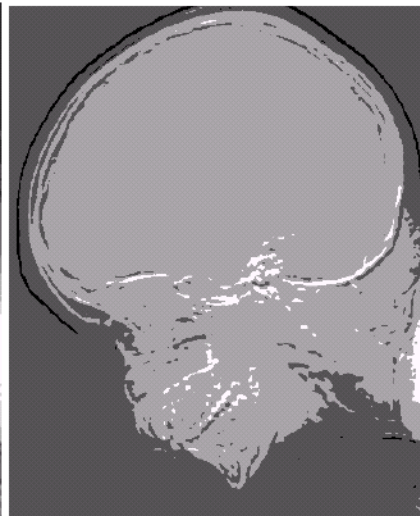
32 grey levels (5 bpp)



16 grey levels (4 bpp)



8 grey levels (3 bpp)



4 grey levels (2 bpp)



2 grey levels (1 bpp)

Resolution: How much is enough?

- ◆ How many samples and gray levels are required for a good approximation?
 - Quality of an image depends on number of pixels and gray-level number
 - The more these parameters are increased, the closer the digitized array approximates the original image
 - But: Storage & processing requirements increase rapidly as a function of N , M , and k

Resolution: How much is enough?

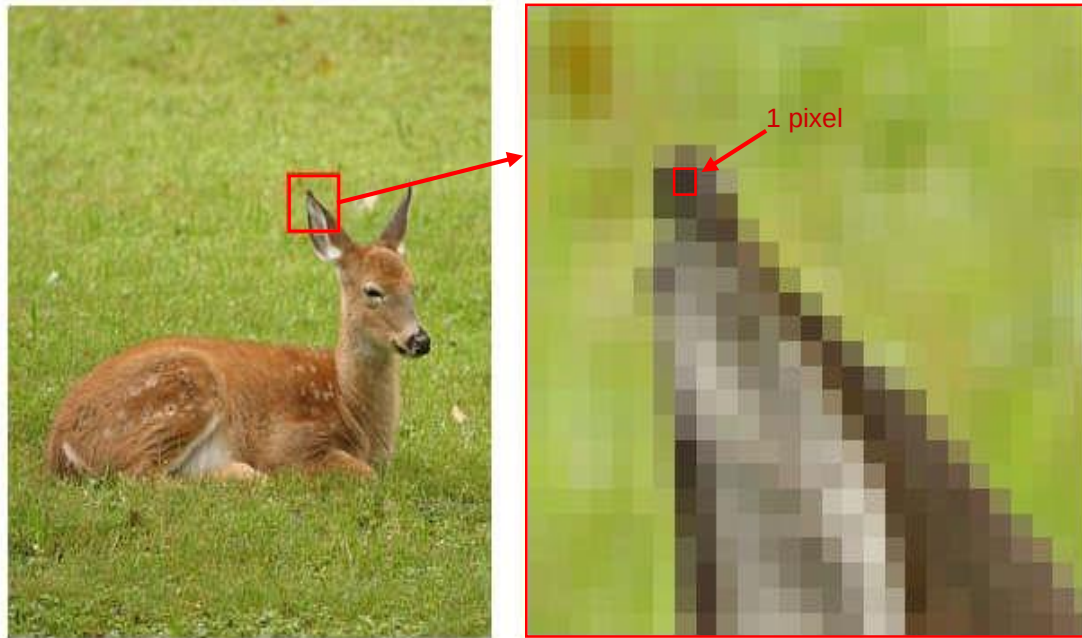
- ◆ Depends on what is in the image and what you would like to do with it



Digital Image

a grid of squares,
each of which
contains a single
color

each square is
called a pixel (for
picture element)

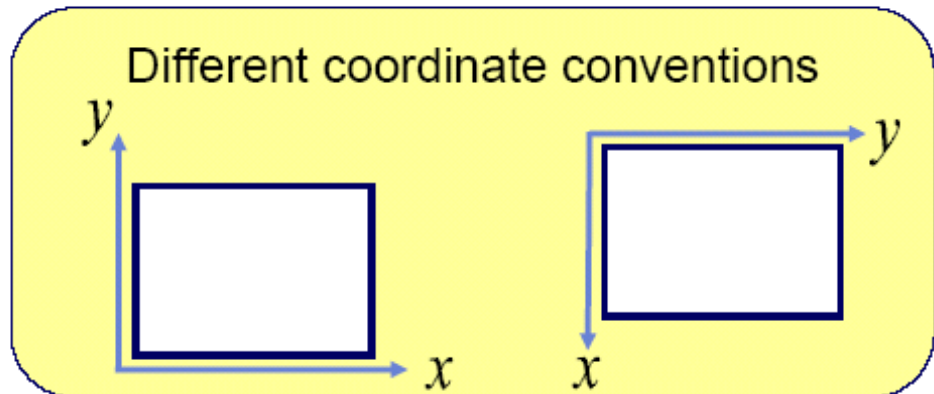
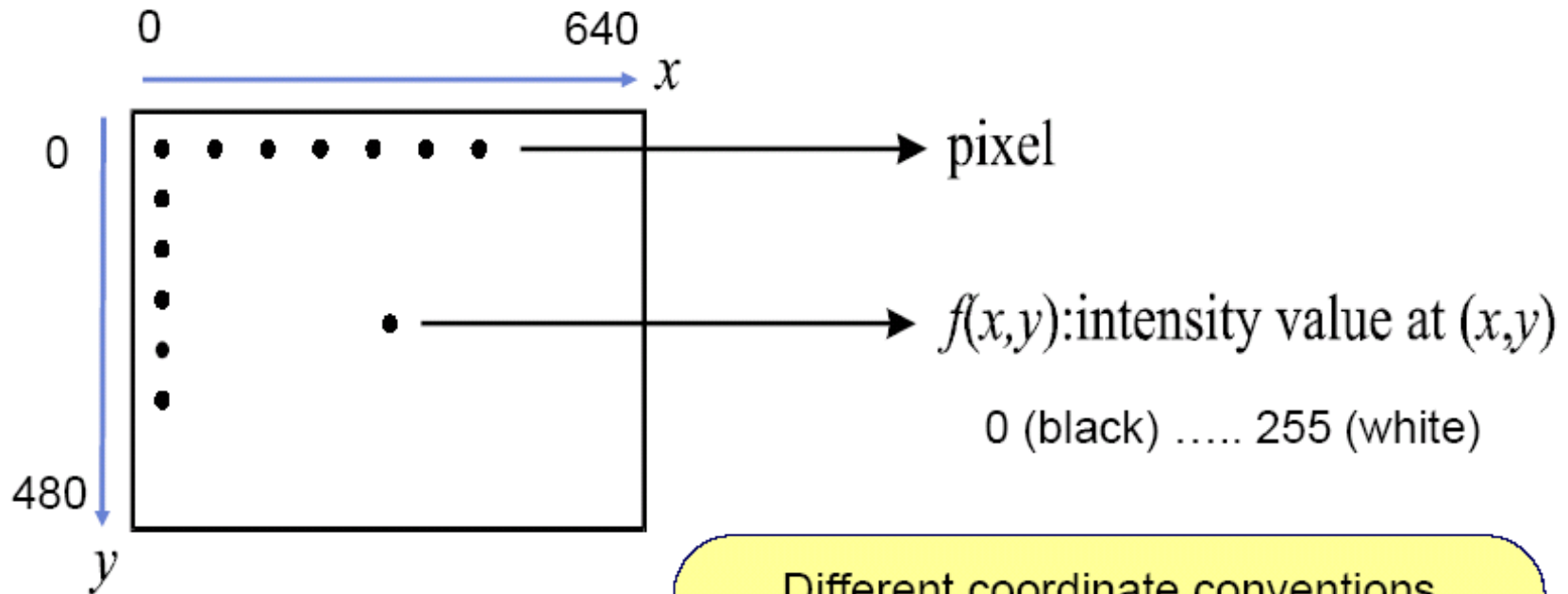


Digital Image

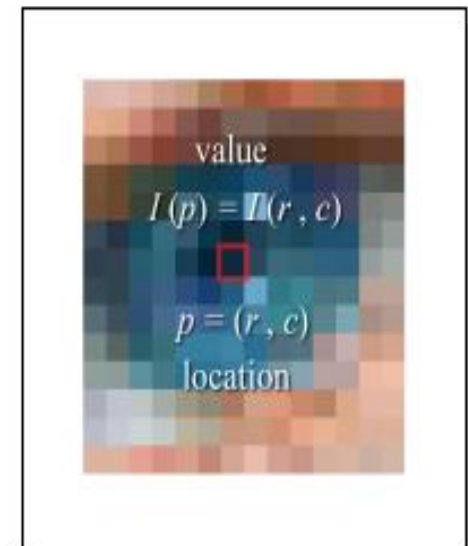
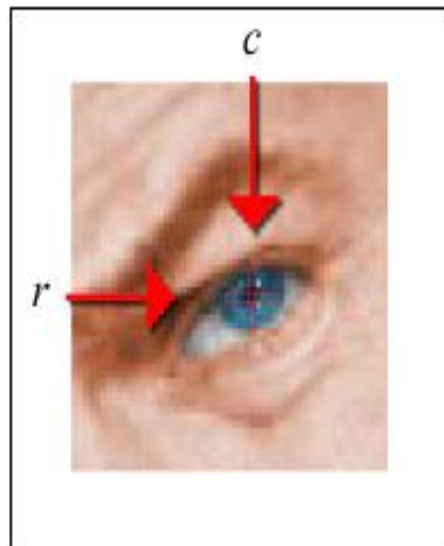
- A set of pixels (picture elements, pels)
- Pixel means
 - pixel coordinate
 - pixel value
 - or both
- Both coordinates and value are discrete

Example

640 x 480 8-bit image



Pixels



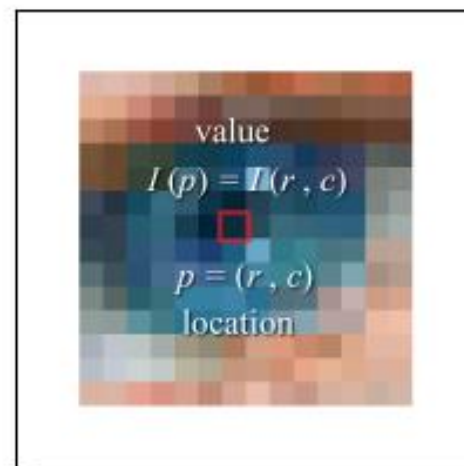
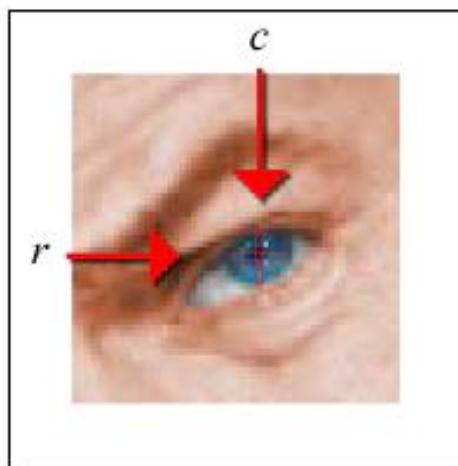
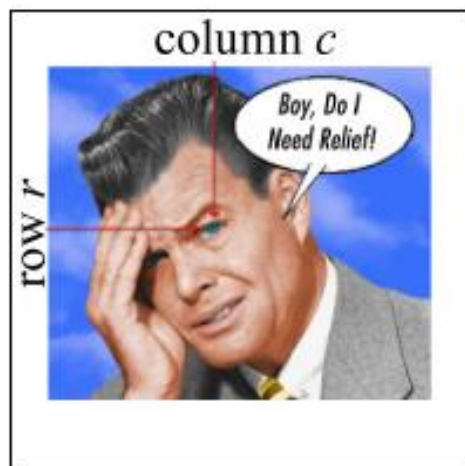
Pixel Location: $p = (r, c)$

Pixel Value: $I(p) = I(r, c)$

Pixel : $[p, I(p)]$

Pixels

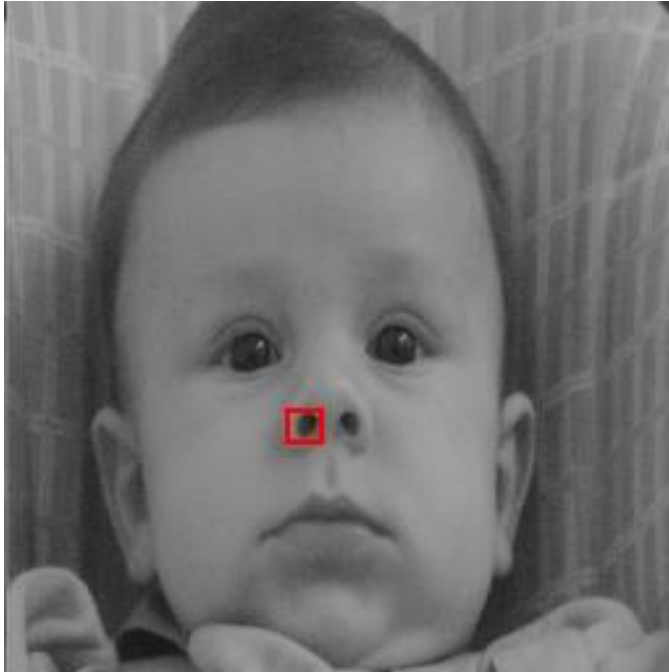
Pixel : $[p, I(p)]$



$$\begin{aligned} p &= (r, c) \\ &= (\text{row \#}, \text{col \#}) \\ &= (272, 277) \end{aligned}$$

$$I(p) = \begin{bmatrix} \text{red} \\ \text{green} \\ \text{blue} \end{bmatrix} = \begin{bmatrix} 12 \\ 43 \\ 61 \end{bmatrix}$$

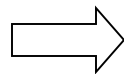
DIGITAL IMAGE REPRESENTATION



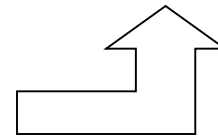
PIXEL VALUES IN HIGHLIGHTED REGION

99	71	61	51	49	40	35	53	86	99
93	74	53	56	48	46	48	72	85	102
101	69	57	53	54	52	64	82	88	101
107	82	64	63	59	60	81	90	93	100
114	93	76	69	72	85	94	99	95	99
117	108	94	92	97	101	100	108	105	99
116	114	109	106	105	108	108	102	107	110
115	113	109	114	111	111	113	108	111	115
110	113	111	109	106	108	110	115	120	122
103	107	106	108	109	114	120	124	124	132

CAMERA



DIGITIZER



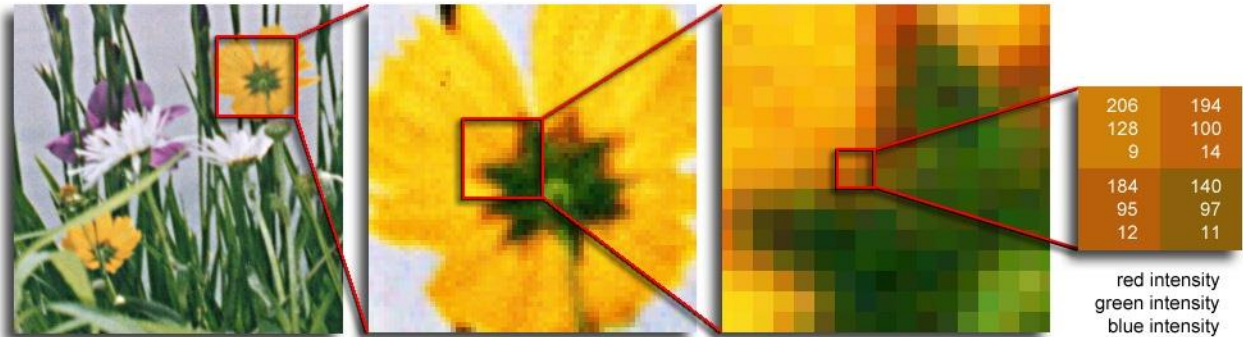
**A set of number
in 2D grid**

Samples the analog data and digitizes it.

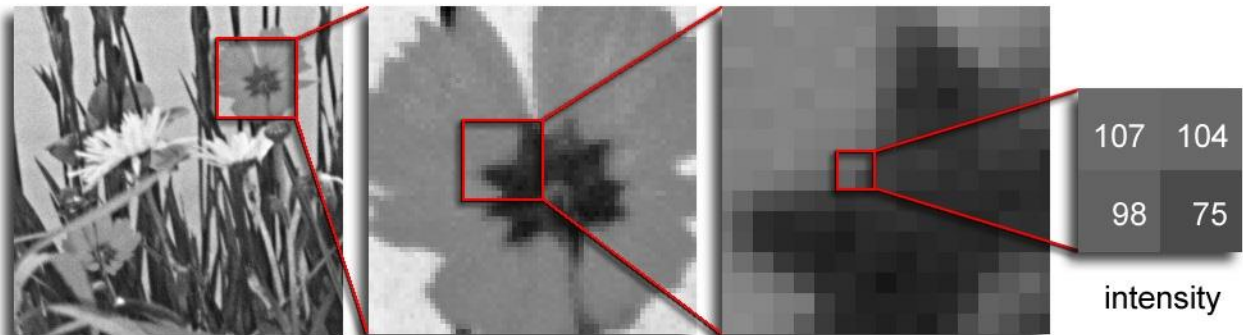
Digital Image

Color images have 3 values per pixel; monochrome images have 1 value per pixel.

a grid of squares, each of which contains a single color



each square is called a pixel (for *picture element*)



What is a Digital Image? (cont...)

- Common image formats include:
 - 1 sample per point (B&W or Grayscale)
 - 3 samples per point (Red, Green, and Blue)



- For most of this course we will focus on grey-scale images

Readings from Book (3rd Edn.)

- Chapter – 1
- Chapter – 2

Read topics from 2.2 to 2.4 from
book



Acknowledgements

- ◆ Statistical Pattern Recognition: A Review – A.K Jain et al., PAMI (22) 2000
- ◆ Pattern Recognition and Analysis Course – A.K. Jain, MSU
- ◆ *Pattern Classification*” by Duda et al., John Wiley & Sons.
- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2008
- ◆ Machine Vision: Automated Visual Inspection and Robot Vision”, David Vernon, Prentice Hall, 1991
- ◆ www.eu.aibo.com/
- ◆ Advances in Human Computer Interaction, Shane Pinder, InTech, Austria, October 2008